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MATE6301
METAL-ORGANIC FRAMEWORKS FOR WATER TREATMENT

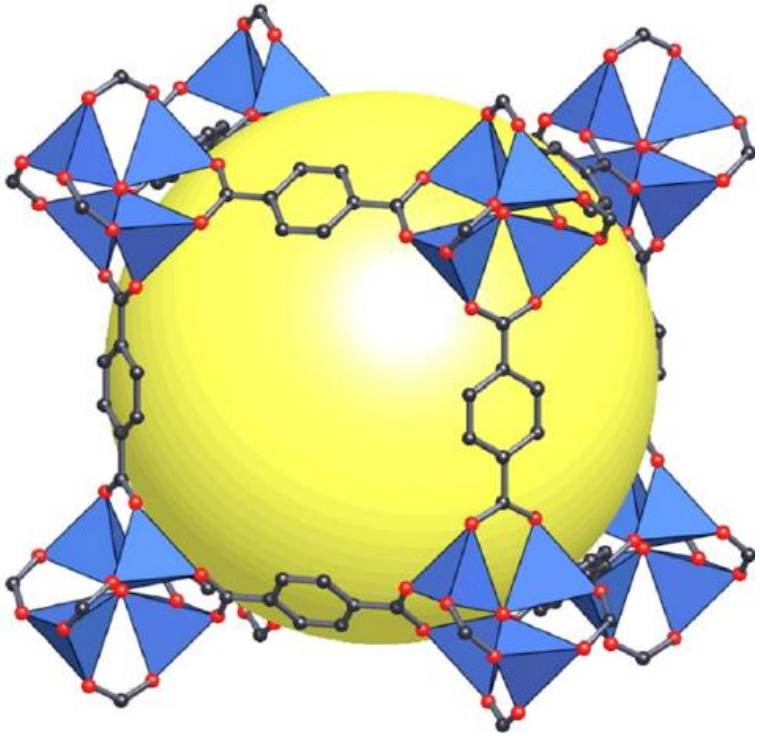
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Australian Institute for Bioengineering and Nanotechnology
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LECTURE OUTLINE

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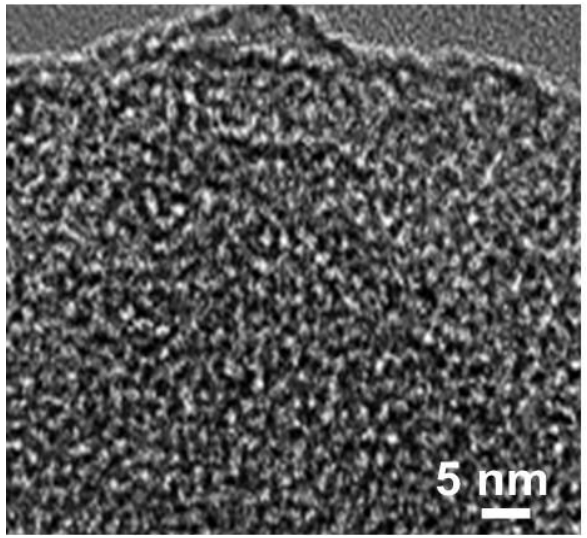
1. Nanoporous Materials
2. Zeolites
3. Metal-organic frameworks (MOFs)
   - Synthesis of MOFs
   - Fabrication of MOF aerogels
   - Characterisation of MOFs
   - Applications of MOFs in water treatment



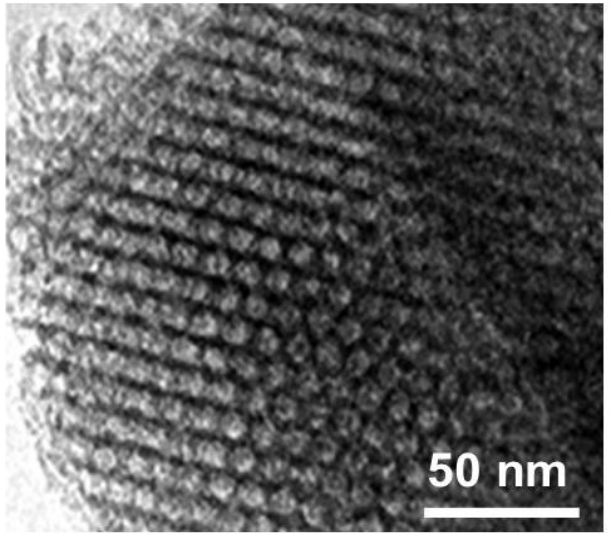
1. NANOPOROUS MATERIALS

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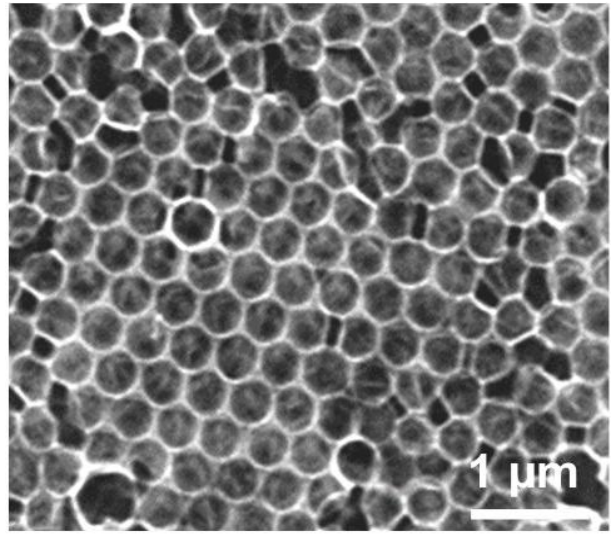
- **Definition:** Porous materials with pore size less than 100 nm.
- Classification of nanoporous materials:



Microporous
(pore size < 2 nm)



Mesoporous
(pore size: 2-50 nm)



Macroporous
(pore size >50 nm)

Figure 1. IUPAC classification of nanoporous materials.

- **Microporous materials:** zeolites, metal-organic frameworks (MOFs)
- **Mesoporous materials:** mesoporous carbon, mesoporous silica, etc.

2. ZEOLITES

- Zeolites are hydrated aluminosilicate minerals made from interlinked tetrahedra of alumina (AlO_4) and silica (SiO_4).

- The general chemical composition of a zeolite is:

$$M^n_x/_n Si_{1-x} Al_x O_2 \cdot yH_2O$$

where M = Na^+, K^+, Li^+, Ca^{2+}, Ba^{2+}…

- Two types of zeolites:

(1) Natural zeolites

- Aluminosilicate minerals
- Occur in nature → volcanic origin
- Over 40 types of natural zeolites
- Commonly used for adsorption, water treatment → very small pore size

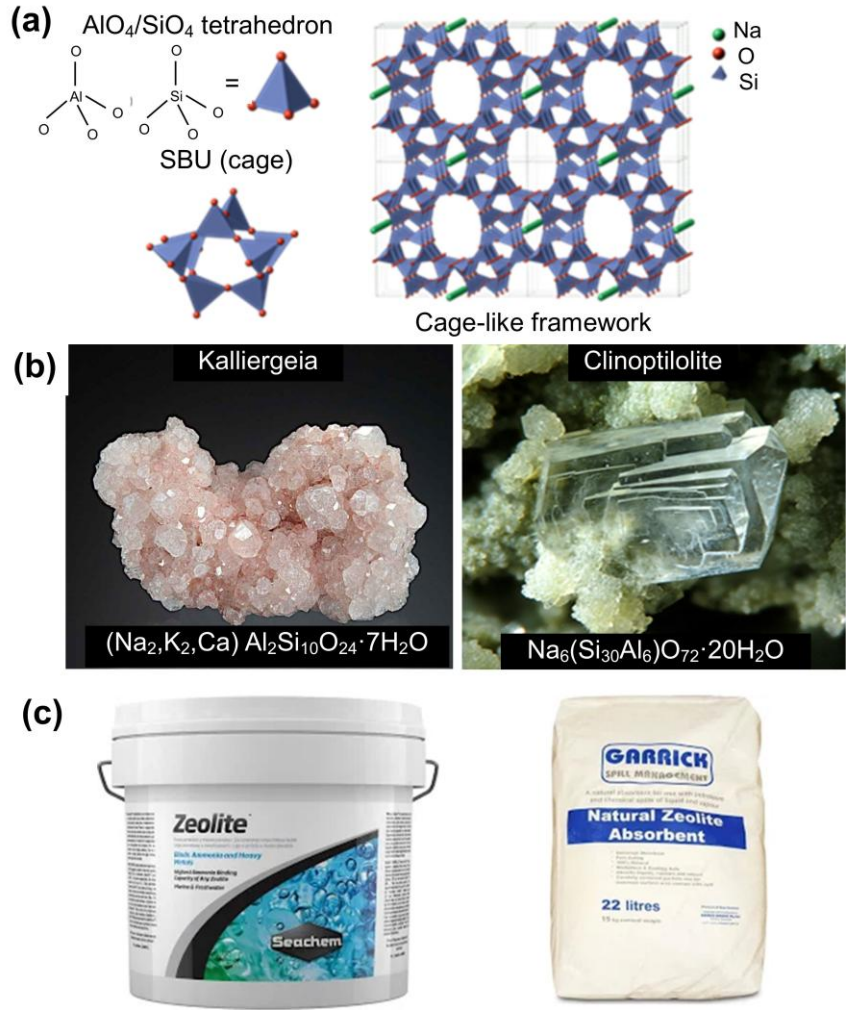


Figure 2. (a) Basic zeolite structure. (b) Examples of natural zeolites, and (c) applications of zeolites.

2. ZEOLITES

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(2) Synthetic zeolites

- Over 100 types of synthetic zeolites.
- Manufactured synthetically using a thermal process.
- Microporous but pore sizes are larger than natural zeolites (~0.2-1 nm).
- Used as catalysts for petroleum cracking, wastewater treatment (heavy metal ion removal) and adsorption.



Figure 3. Examples of synthetic zeolite catalysts (ZSM-5 & 13A).

3. METAL-ORGANIC FRAMEWORKS (MOFs)

- **Definition**: Crystalline, porous materials formed by the simple coordination of metal ions and organic linkers (ligands).
- Pioneered in late 90's by Professor Omar Yaghi from UC Berkeley (USA) and Professor Susumu Kitagawa from Kyoto University (Japan)
- Since late 90's, >90,000 MOFs have been synthesized and over 500,000 MOFs have been predicted by simulations.

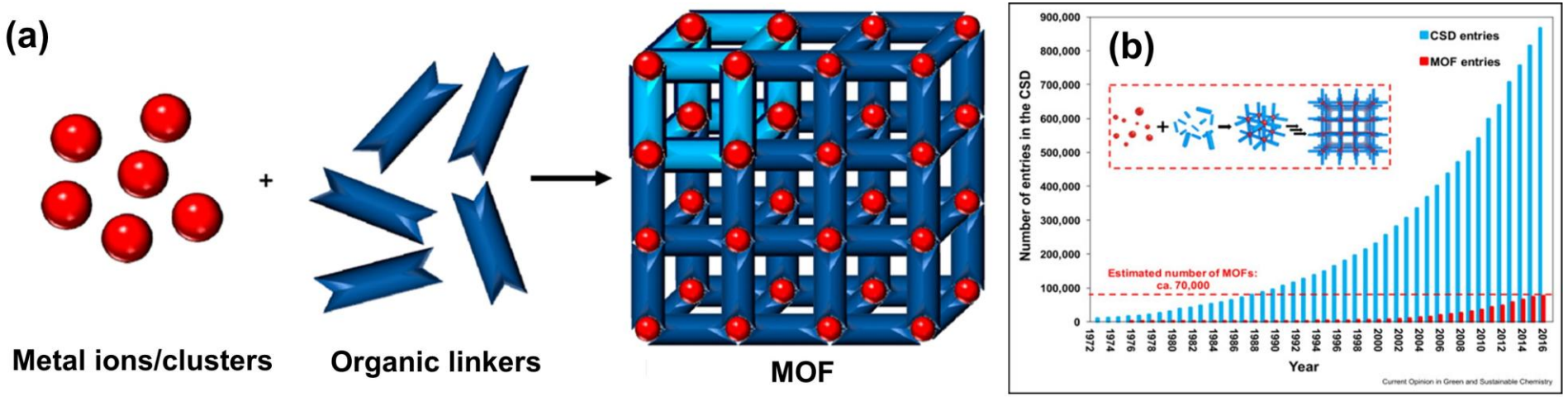


Figure 4. (a) Schematic illustration showing the general assembly process of MOF. (b) A plot showing the number of MOFs listed in Cambridge Crystal Database from 1972 to 2016.

3.1. COMPOSITIONS OF MOFs

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- MOFs have controllable composition. They can be prepared by combining different metal ions (centers) and organic linkers.

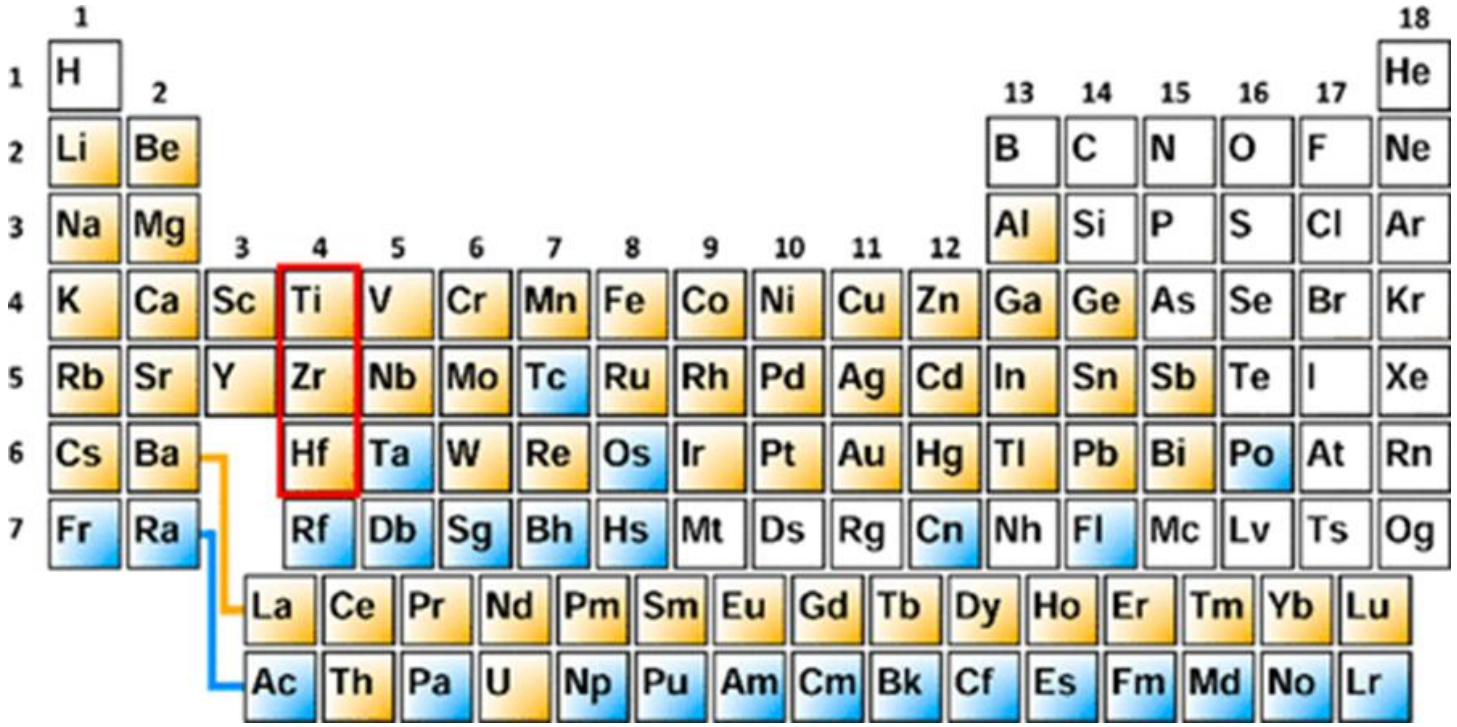


Figure 5. Various metal centres used for the preparation of MOFs.

- Metal centres in MOFs:

Divalent metals: Ni^{2+}, Co^{2+}, Mn^{2+}, Zn^{2+}, Cu^{2+}

Trivalent metals: Al^{3+}, Cr^{3+}, In^{3+}

Tetravalent metals: Ti^{4+}, Zr^{4+}, Sn^{4+}

Higher valency metals: W^{6+}

3.2. ORGANIC LINKERS FOR MOFs

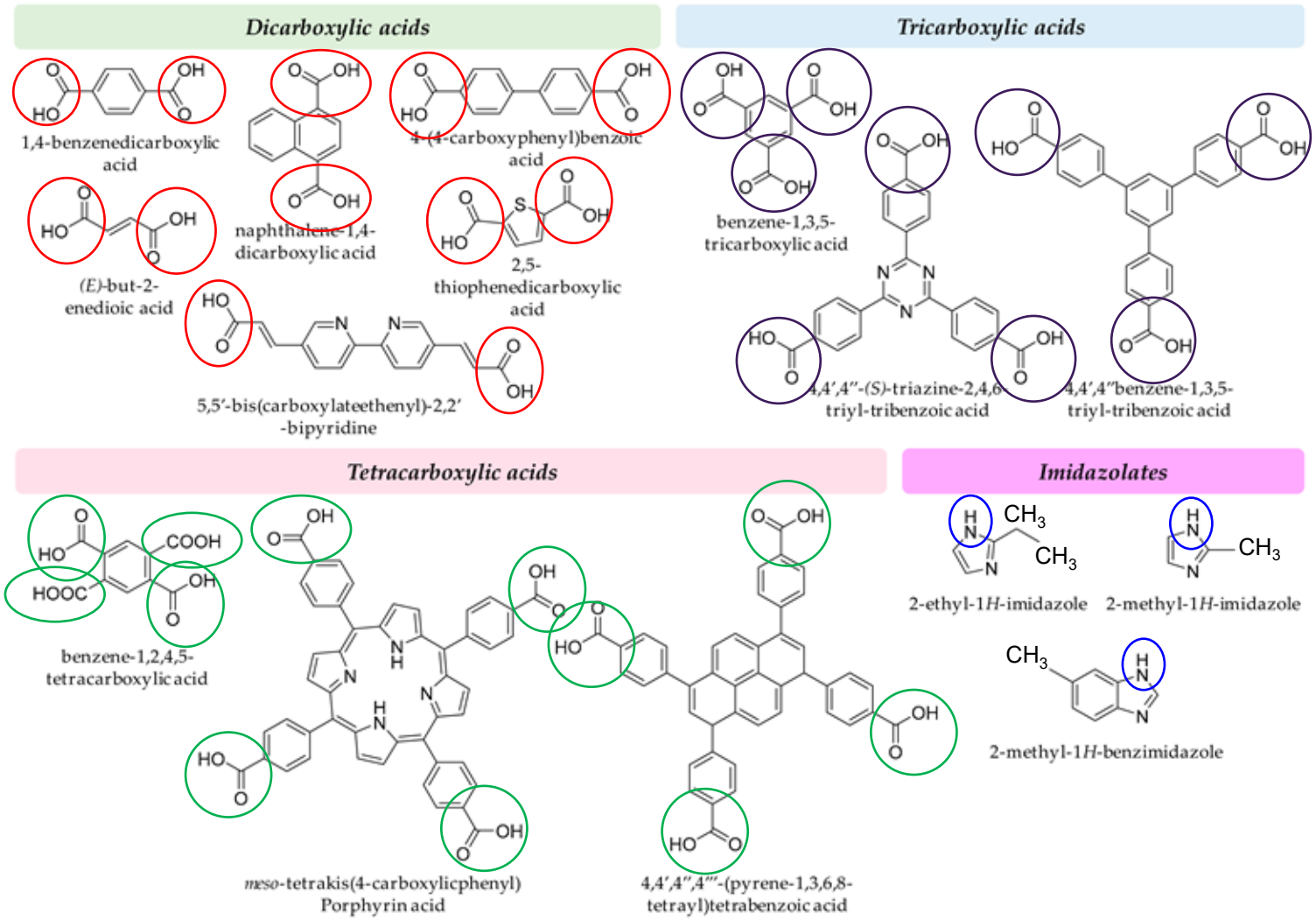


Figure 6. Common linkers used for the preparation of metal-organic frameworks (MOFs).

Typical linkers in MOFs:

- **Dicarboxylic acids:** contain two carboxylic acid (-COOH) groups, e.g., 1,4-benzenedicarboxylic acid (often shortened as “BDC”)
- **Tricarboxylic acids:** contain three carboxylic acid groups, e.g., benzene-1,3,5-tricarboxylic acid (often shortened as “BTC”)
- **Tetracarboxylic acids:** contain four carboxylic acid groups, e.g., benzene-1,2,4,5-tetracarboxylic acid
- **Imidazolates:** contain two non-adjacent nitrogen atoms, e.g., 2-methylimidazole (2-MeIM)

3.3. EXAMPLES OF MOFs

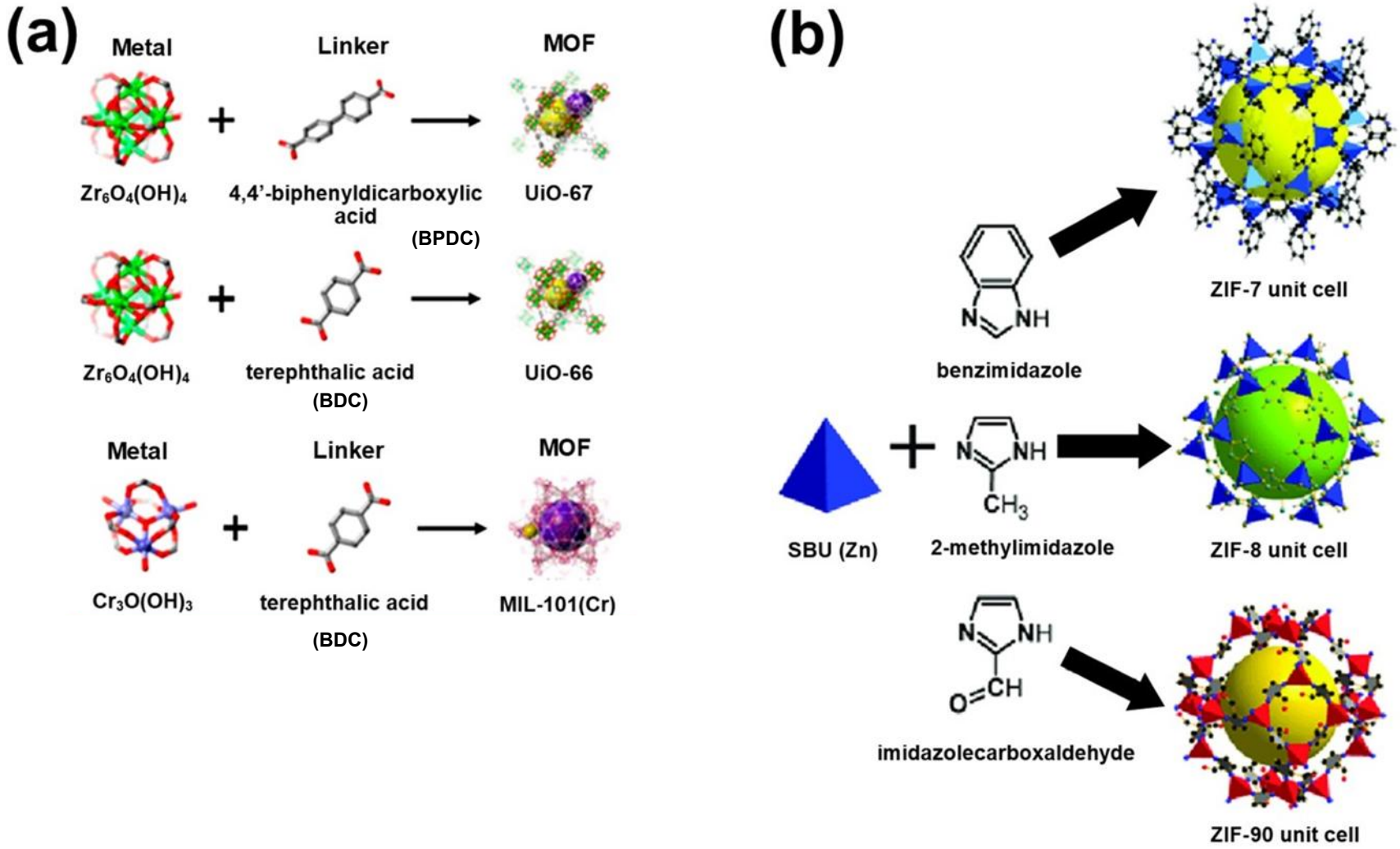
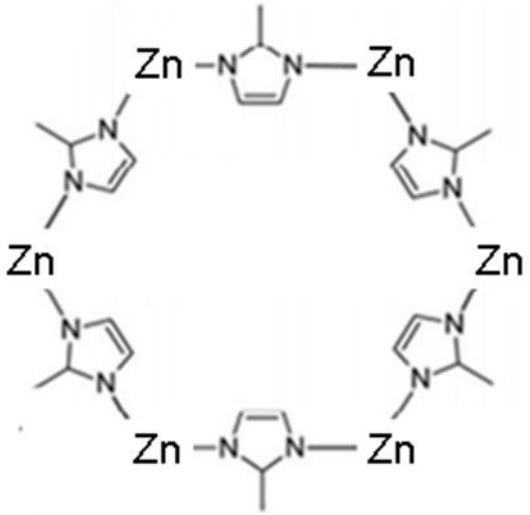


Figure 7. (a) Examples of different MOFs obtained by combining different metal centres and organic linkers. (b) Examples of different ZIF-based MOFs which can be obtained using Zn metal centre and different imidazolate-based linkers.

Zeolitic imidazolate frameworks (ZIFs)

- ZIF-8 (ZIF-67) is formed by coordinating zinc (cobalt) ions with 2-methylimidazole linker.
- Exhibit similar structu conventional alumino
- The tetrahedral Si or O in zeolites are repla metals (e.g., Zn or Co linkers, respectively.
- Properties: microporous, high surface area, good thermal stability, decent chemical stability, high crystallinity, etc.



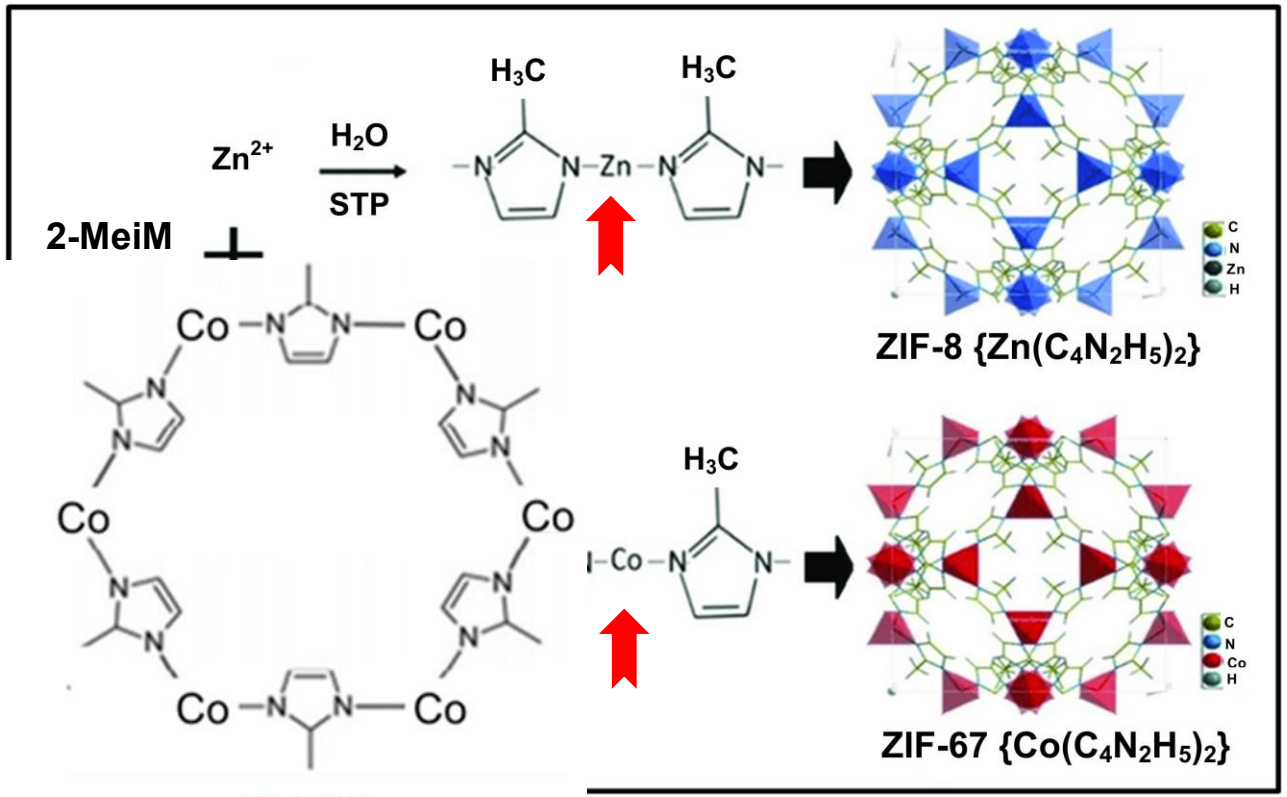


Figure 8. Schematic illustration showing the formation of ZIF-8 and ZIF-67 from zinc (cobalt) ions and 2-methylimidazole. Inset shows the typical morphology of ZIF-8 or ZIF-67 particles.

3.5. PROPERTIES OF MOFs

- Controllable composition, structure, and porosity.
- Generally microporous.
- High internal surface area (up to 8,000 m^2/g).
- High thermal stability (300-600 ºC in air).
- Low thermal conductivity. (<2 $W\ m^{-1}\ K^{-1}$ at RT ≈ concrete)
- Generally low electrical conductivity.

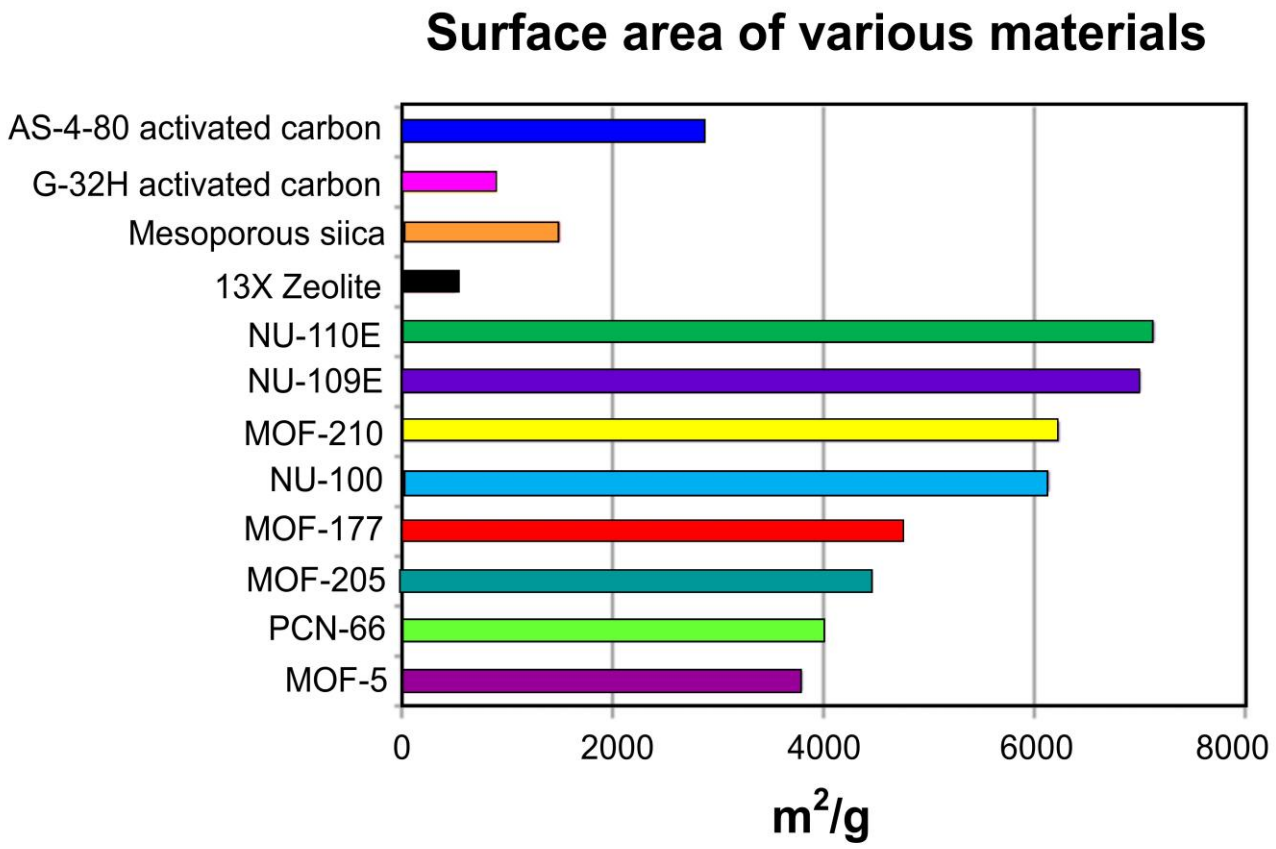


Figure 9. Comparison of surface areas of MOFs and other porous materials.

3.5. PROPERTIES OF MOFs vs. ZEOLITES

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Metal-organic frameworks	Zeolites
Higher surface area (up to 8,000 m^2/g)	Lower surface area (500-800 m^2/g)
Highly controllable composition	Limited control of composition
Controllable pore size (micro, meso, hierarchical pores)	Limited pore size (microporous, 0.2-1 nm)
Lower thermal stability (300-600 ºC)	Higher thermal stability (up to 800 ºC)
Lower chemical stability	Higher chemical stability
Higher cost (some commercialised)	Lower cost (many commercialised)

- Introduction to MOFs: https://www.youtube.com/watch?v=m91P-R3kxOs

3.6. SYNTHESIS OF MOFs

(1) Precipitation

- Aging at low temperature (RT to <100 ºC).
- Performed at ambient pressure.
- Used to synthesize MOFs that readily precipitate at low temperatures
  e.g., ZIF-8, ZIF-67, HKUST-1 (Cu-BTC)

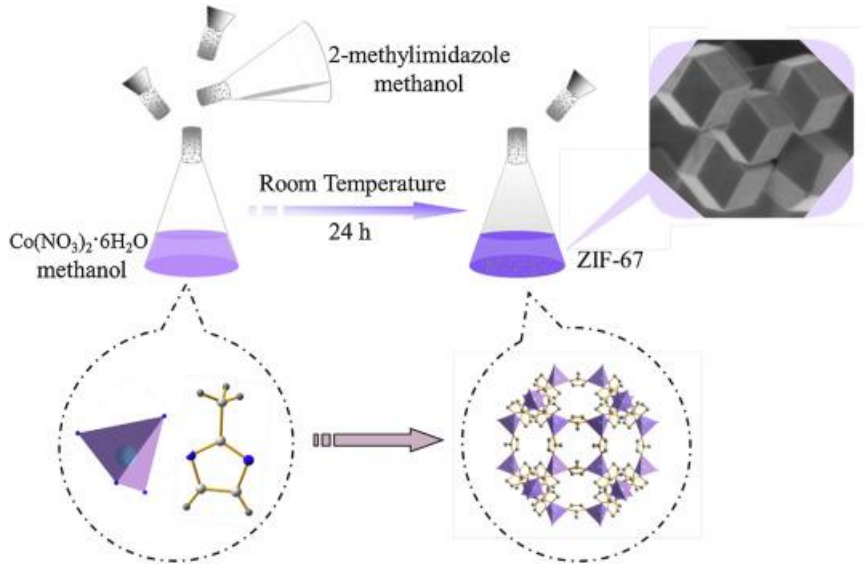


Figure 10. Synthesis of ZIF-67 particles by precipitation method.

(2) Hydrothermal/solvothermal method

- *Hydrothermal*: uses aqueous media
- *Solvothermal*: uses non-aqueous media
- Conducted at high temperature and high pressure.
- Performed in stainless-steel autoclaves.
- Examples: ZIF-8, ZIF-67, MIL-125 (Ti-BDC)

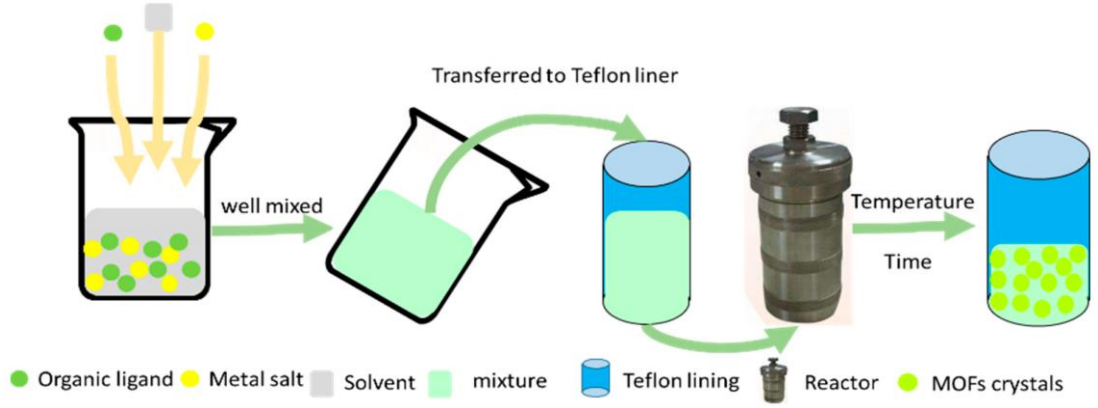


Figure 11. General synthesis of MOF particles by hydro-/solvothermal method.

3.6. SYNTHESIS OF MOFs

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(3) Microwave method

- More homogenous than hydro-/solvothermal method.
- Fast reaction kinetics.
- High product yield.
- Examples: ZIF-8, UiO-66 (Zr-BDC)

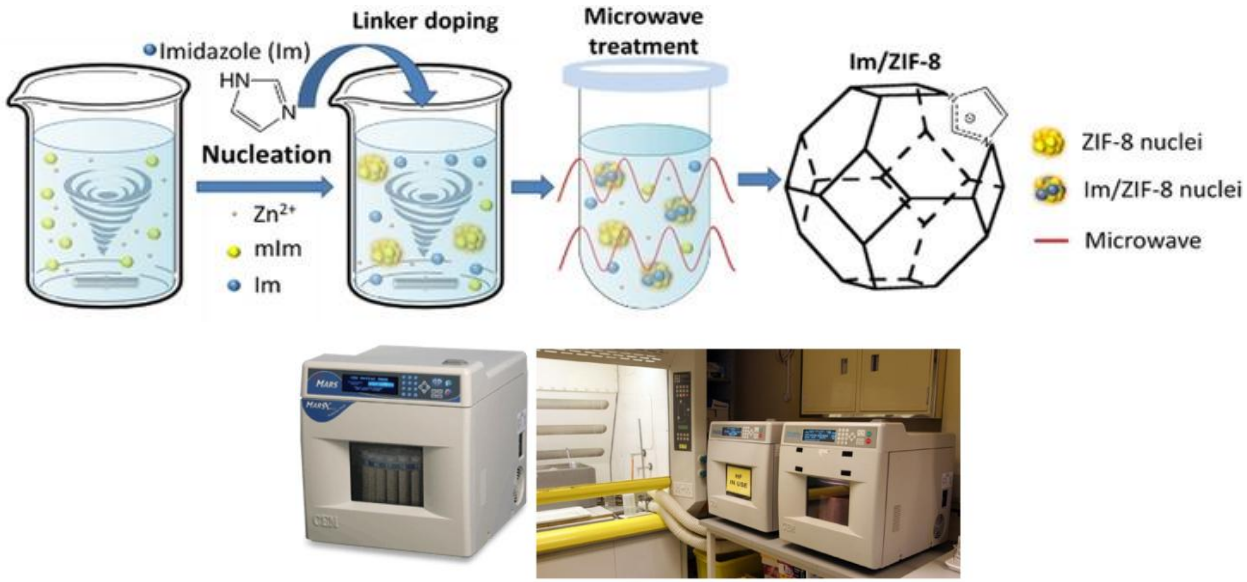


Figure 12. Microwave synthesis of ZIF-8 particles.

(4) Electrochemical method

- Metal ions are supplied from anodic dissolution (instead of metal salts) to react with dissolved linker molecules.
- Cu cathode is used & metal deposition is avoided by using protic solvents.
- Rapid, low temperature, and separation of metal ions are not needed.
- Examples: ZIF-8, ZIF-67, HKUST-1

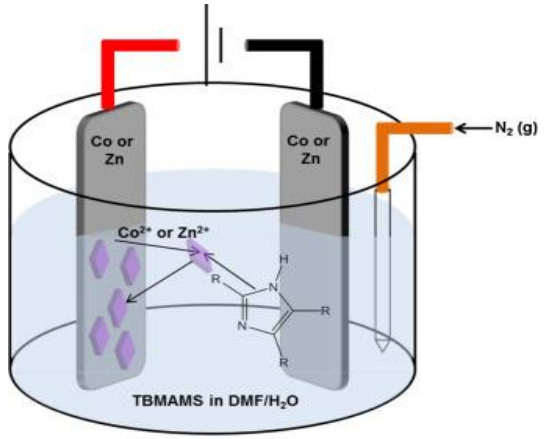


Figure 13. Electrochemical synthesis of ZIF particles.

3.6. SYNTHESIS OF MOFs

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(5) Sonochemical method

- Involves formation and collapse of bubbles ("acoustic cavitation").
- Very high local T (~4727 ºC) & P (~1,000 bar).
- Very fast heating & cooling rates (>734 ºC/s).
- Produce fine crystallites.

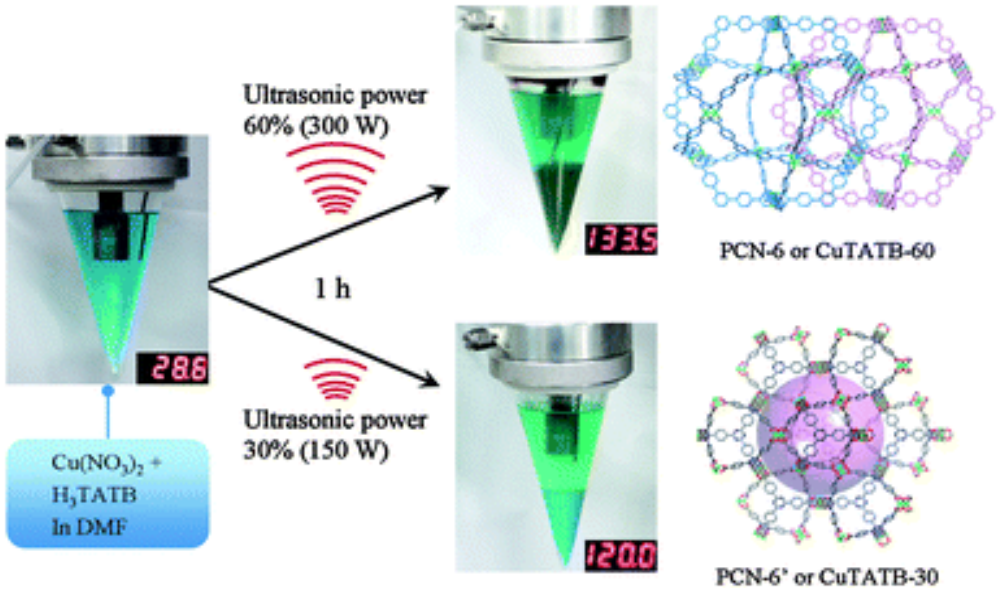


Figure 14. Synthesis of MOF particles by sonochemical method.

(6) Mechanochemical method

- Does not require solvents.
- Good for scalable production.
- Generates nanosized MOFs.
- Performed at room temperature using ball milling or mortar and pestle.
- Structural control can be challenging.



Figure 15. Synthesis of ZIF-8 and ZIF-67 particles by ball-milling method.

3.7. MOF CHARACTERISATION

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XRD



FTIR



BET



EM



Figure 16. Different techniques used for characterizing the structural and compositional features of MOFs.

- Composition
- Crystal structure

- Composition (organic compounds, bonds: C−C, C−H, C=O, −COOH, etc.)

- Surface area
- Pore size distribution

- Exterior morphology
- Pores (mesopores and macropores)
- Composition (EDS)

3.7. MOF CHARACTERISATION –XRD & EM

XRD

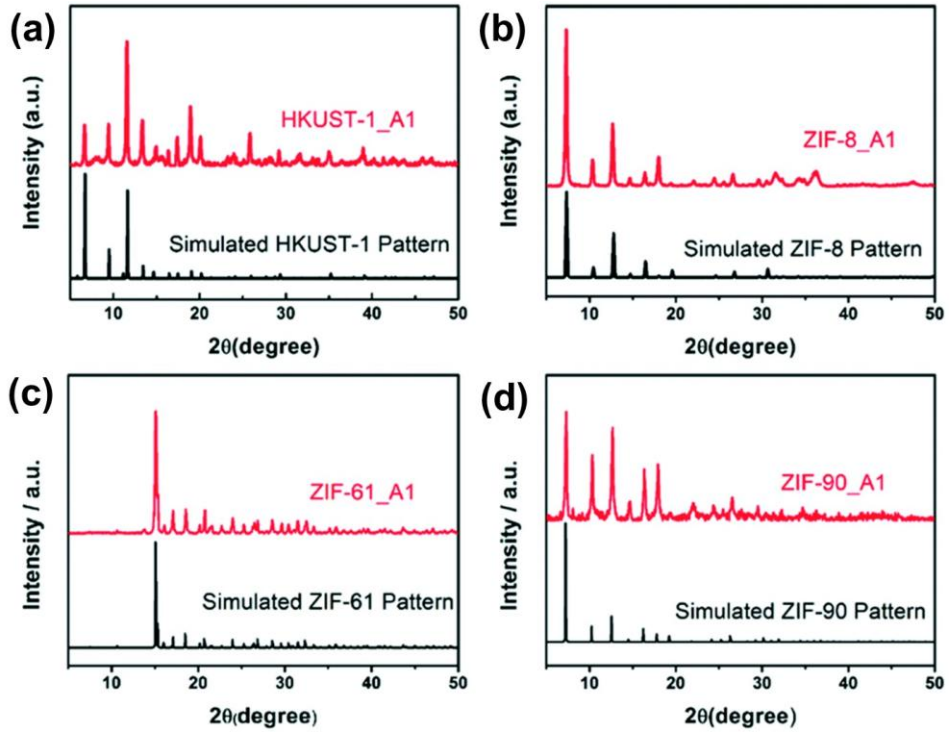


Figure 17. XRD patterns of (a) HKUST-1, (b) ZIF-8, (c) ZIF-61, and (d) ZIF-90

- Highly crystalline.
- Match experimental pattern with simulated pattern (CCDC Database).

SEM & TEM – ZIF-67

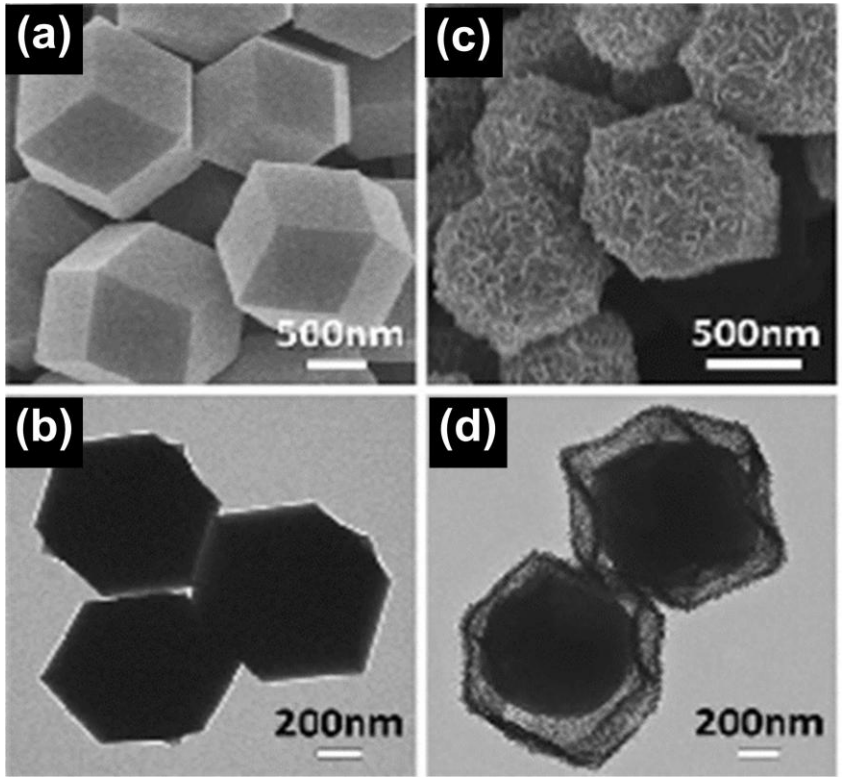


Figure 18. SEM and TEM images of (a, b) solid and (c, d) hollow ZIF-67 particles.

- **SEM:**
- **TEM:**

3.7. MOF CHARACTERISATION-FTIR & BET

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FTIR

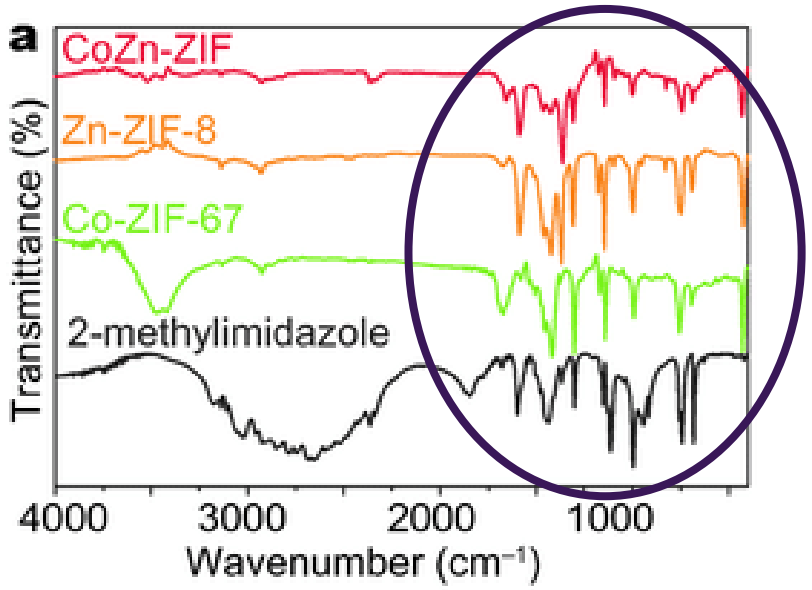


Figure 19. FTIR spectra of 2-MeIM and ZIF particles.

- **FTIR:** information about organic bonds present in the samples.
- Check similarities of MOF peaks with the linker molecules to confirm successful formation of MOFs.

BET

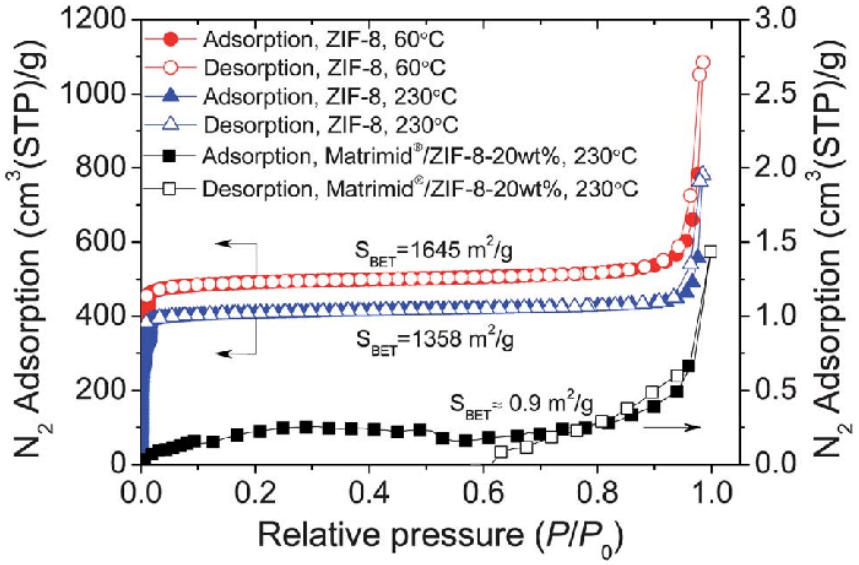


Figure 20. N_2 sorption isotherms of ZIF particles.

- **BET** (Brunauer-Emmett-Teller) surface area: information about surface area and pore size distribution.
- Measures volume of N_2 adsorbed/desorbed.

3.7. MOF CHARACTERISATION-BET S.A.

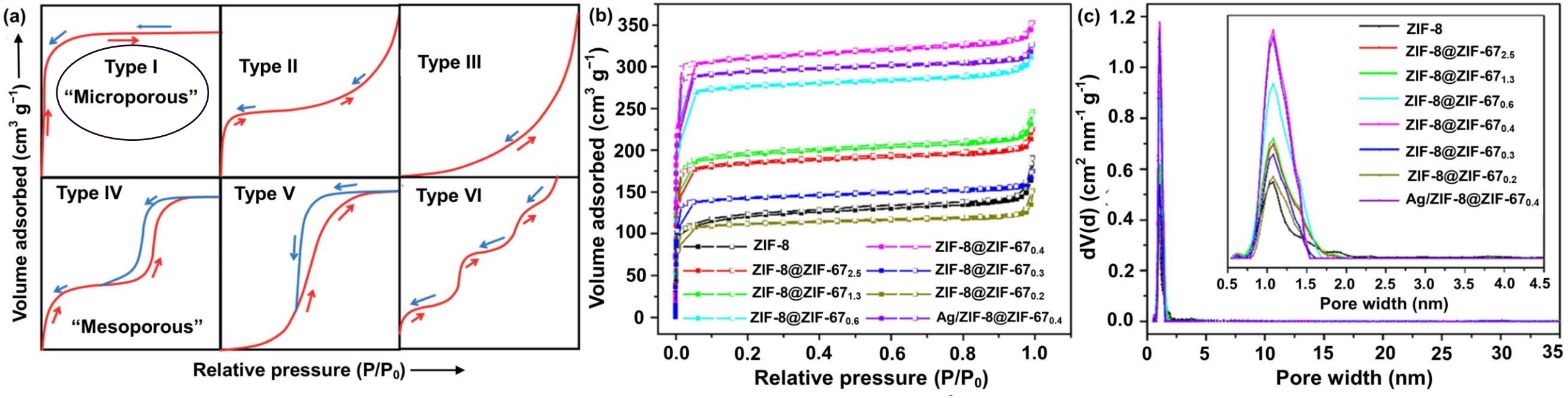


Figure 21. (a) Types of sorption isotherms. (b) N_2 sorption isotherms and (c) pore size distribution plots of ZIF-8 and ZIF-8@ZIF-67 composites.

- The N_2 sorption isotherms provide information about the type of pores present in the material, while the pore size distribution data gives information about the size distribution of pores inside the MOFs.

- MOFs typically exhibit type I isotherm due to their microporosity (pore size< 2.0 nm).

3.8. SIZE CONTROL OF MOFs

- The formation of MOFs can be divided into several stages:

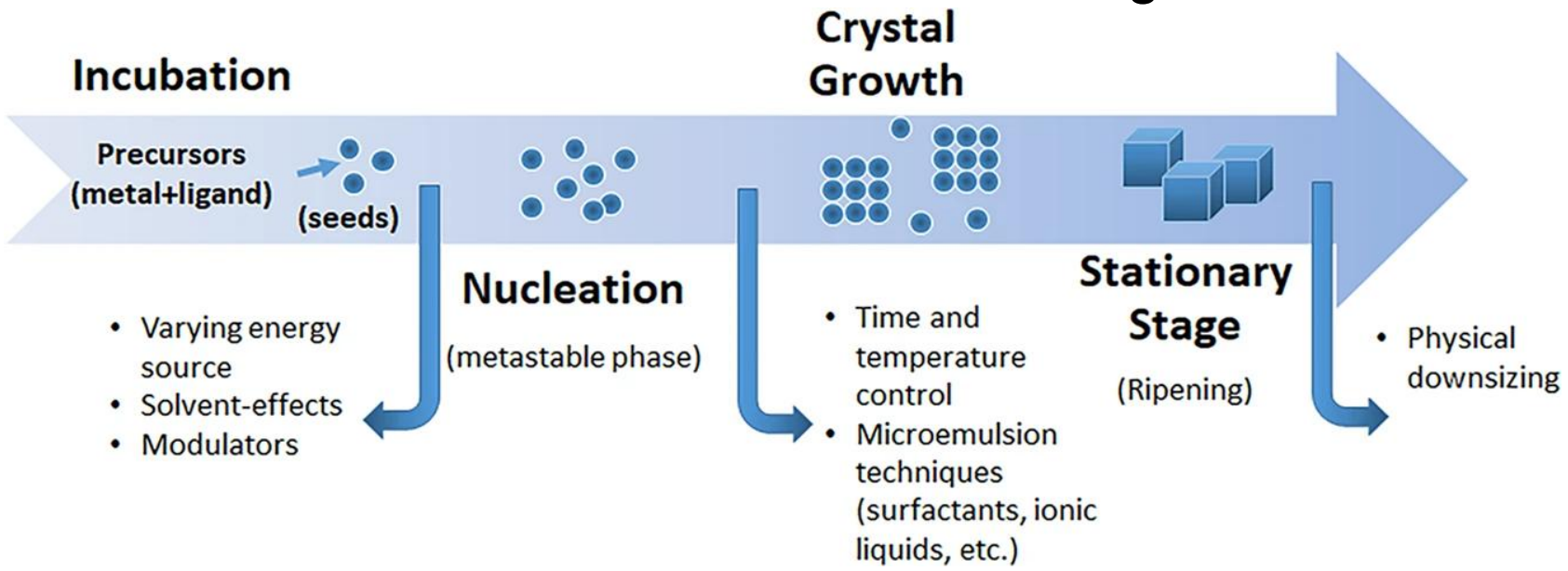


Figure 22. Major stages of MOF crystallization.

- MOF particle size is determined by the nucleation and growth rates.
- Faster nucleation → more nuclei → shorter crystal growth → smaller particles
- Slower nucleation → less nuclei → longer crystal growth → larger particles
- **Why control the size of MOFs?**
  (1) Increase surface area → increase the number of active sites
  (2) Improve ion/electron transfer and reduce ion diffusion length

3.8. SIZE CONTROL OF MOFs: ZIF-8

- Size control of ZIF-8 particles can be achieved by modifying:

(1) The anion reactivity of metal salts: more reactive anions → faster nucleation → more nuclei → shorter crystal growth → smaller particles.

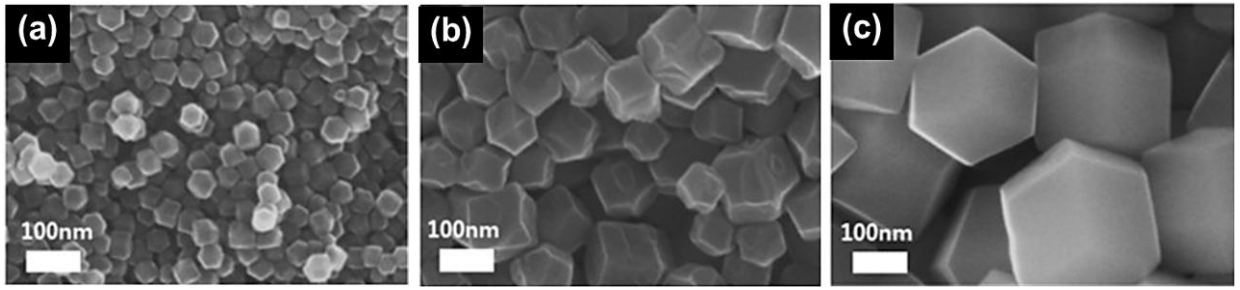


Figure 23. SEM images of ZIF-8 particles obtained by mixing (a) zinc nitrate (88 nm), (b) zinc acetate (240 nm), and (c) zinc chloride (533 nm) and 2-methylimidazole at 70 ºC for 10 mins.

$$NO_3^- > CH_3COO^- > Cl^-$$

(2) Metal to linker ratio: Higher L:M ratio (i.e., excess linker) → faster nucleation → more nuclei → shorter crystal growth → smaller particles.

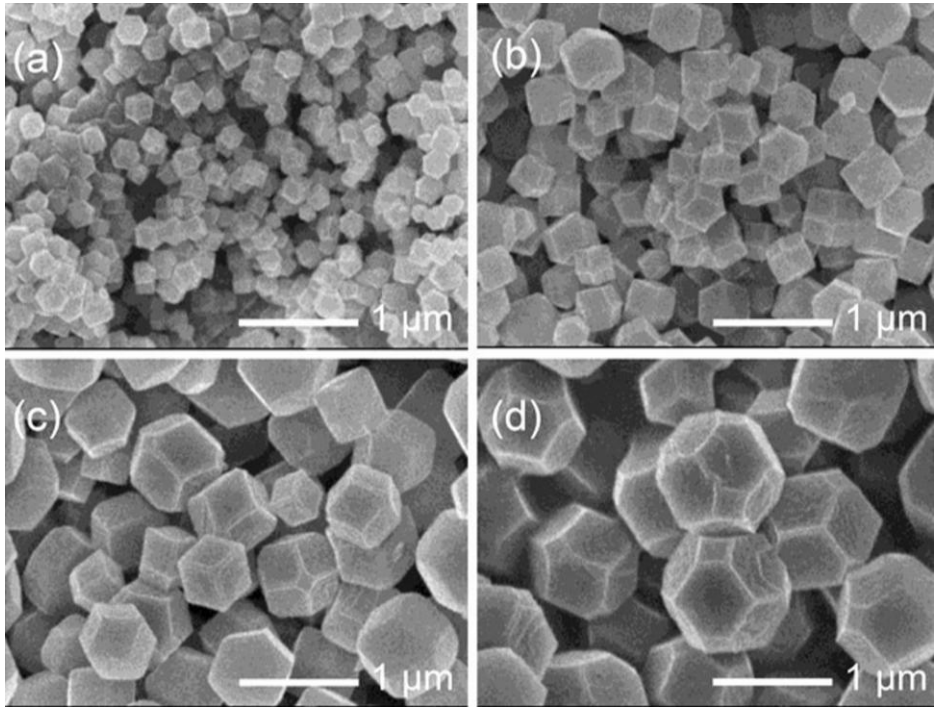


Figure 24. SEM images of ZIF-8 particles obtained with 2-MeIM: Zn^{2+} ratios of (a) 40:1, (b) 30:1, (c) 20:1, and (d) 10:1.

3.8. SIZE CONTROL OF MOFs: ZIF-8

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(3) Reaction temperature: higher temperature → faster nucleation → more nuclei → shorter crystal growth → smaller particles.

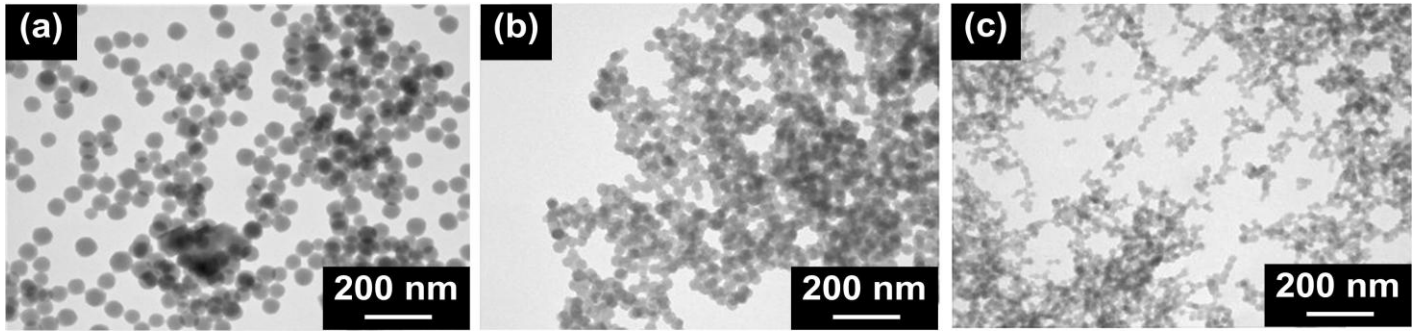


Figure 25. SEM images of ZIF-8 particles obtained at reaction temperatures of (a) -15 ºC, (b) 25 ºC, and (c) 60 ºC in methanol (Zn^{2+}:2-MeIM = 1:8)

(4) Reaction time: longer time → longer crystal growth → larger particles

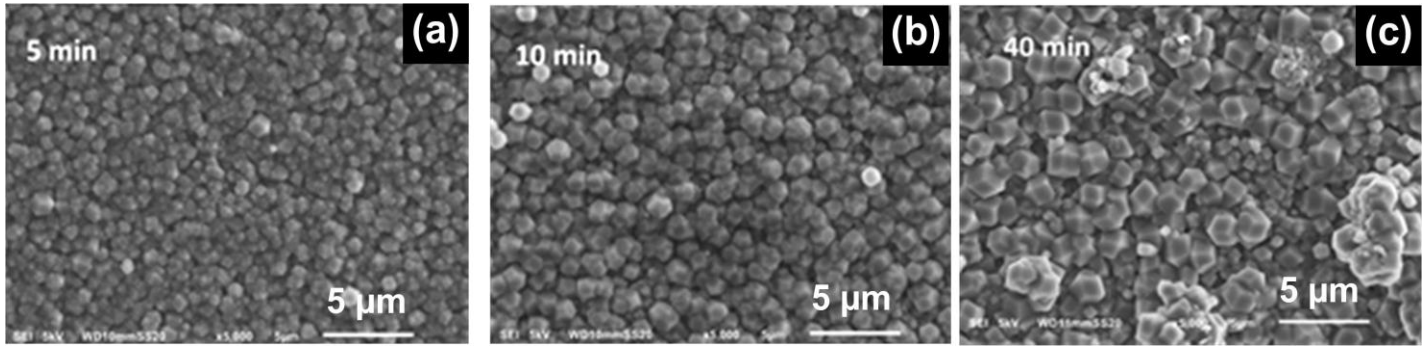


Figure 26. SEM images of ZIF-8 particles obtained with reaction time of (a) 5 mins, (b) 10 mins, and (c) 40 mins using zinc oxide rods and 2 M 2-MeIM.

3.10. MOF AEROGELS

- MOFs are typically produced in powder forms.

- Limitations of MOF powders for industrial applications:
  - Low processibility
  - Difficult recovery
  - Potential of causing secondary/additional pollution

- Enhancing processibility of MOFs → by forming aerogels.

- What is an aerogel?
  "**Aerogel** is a synthetic porous ultralight material obtained by combining a polymer with a solvent to form a gel, and then removing the liquid from the gel."

- **Characteristics of aerogels**: very light, highly porous, high surface area, low density, low thermal conductivity.

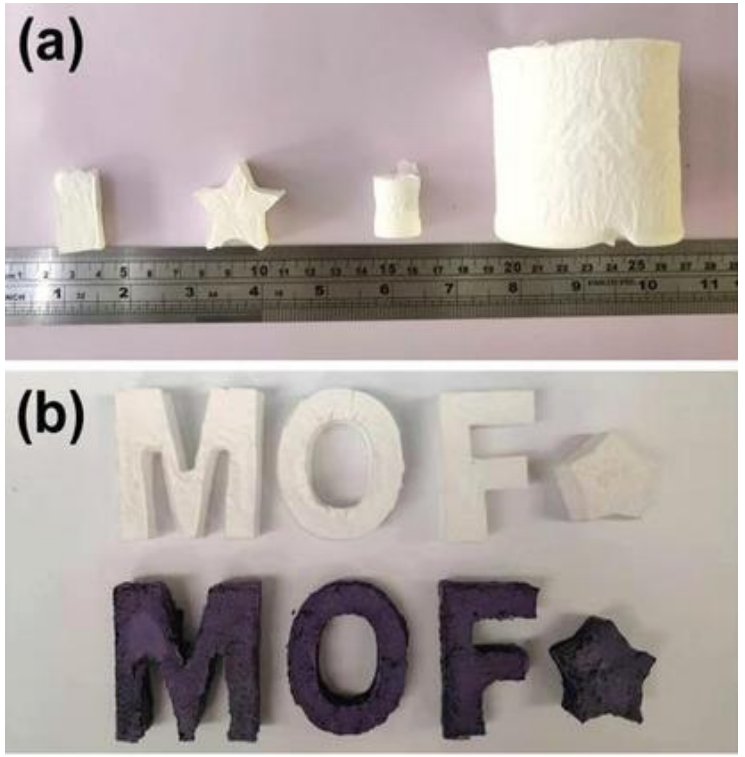


Figure 31. (a, b) Digital photographs of ZIF-8 and ZIF-67 aerogels.

3.10. MOF AEROGELS

Two general methods for fabricating MOF aerogels:

(1) **Mixing method**: Pre-synthesized MOF particles are mixed directly with the supporting matrix with heating to form an aerogel.

(2) **In situ growth:** MOF particles are grown directly on the supporting matrix to form an aerogel. This method is superior as it allows for more uniform distribution of the MOF particles on the supporting matrix.

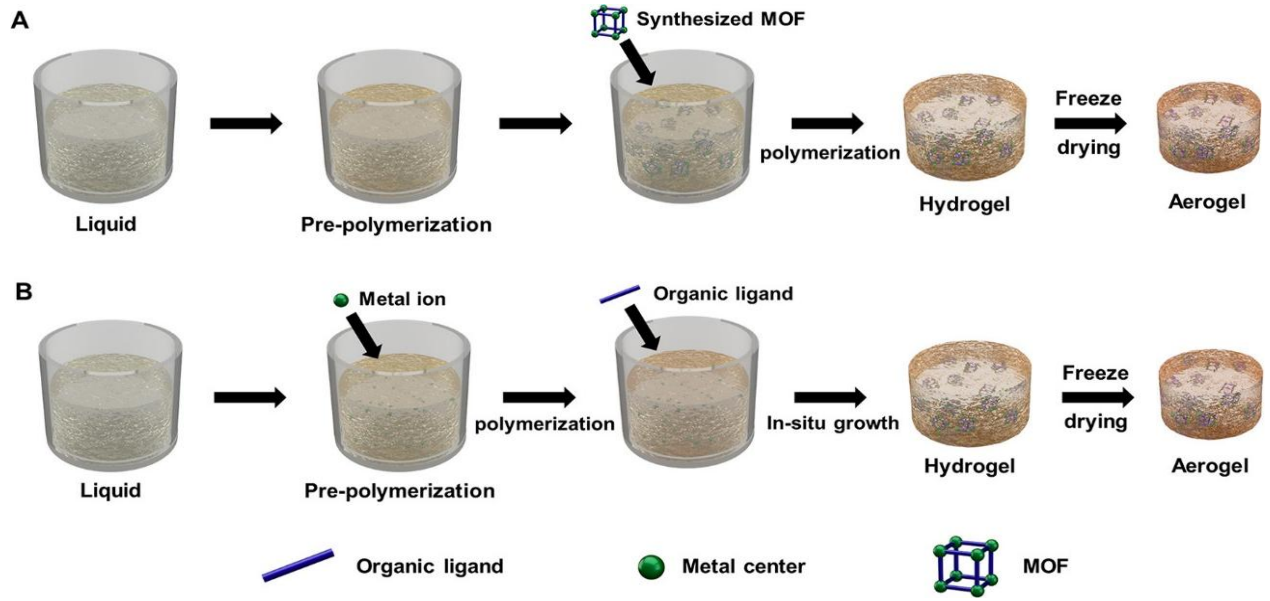


Figure 32. Schematic illustrations showing the preparation of aerogel by (a) mixing method and (b) in-situ growth method.

3.10. MOF AEROGELS- MIXING METHOD

- ZIF-8 aerogel can be prepared by mixing ZIF-8 powder with agarose (as a supporting matrix to promote crosslinking) and subsequent heating at mild temperatures to prepare the ZIF-8 hydrogel.

- Freeze drying is applied to remove water from the hydrogel and generate ultralight ZIF-8 aerogel.

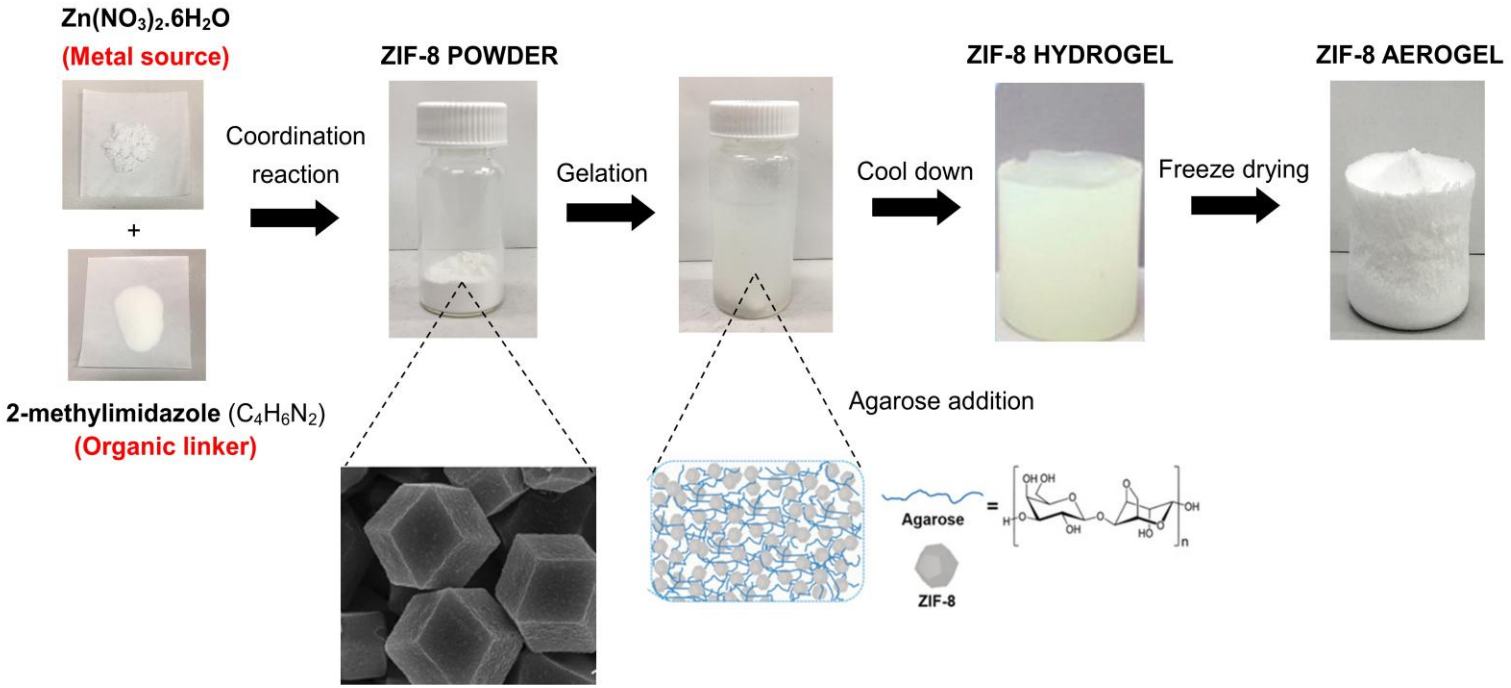


Figure 33. Preparation process of ZIF-8 aerogel by mixing with agarose.

3.10. MOF AEROGELS- IN-SITU METHOD

- Cellulose aerogel (CA) derived from cotton pulp using epichlorohydrin as a crosslinking agent.
- ZIF-67 particles can be directly grown ("in-situ" growth) on the CA aerogel by dispersing it in a methanolic solution containing cobalt nitrate and 2-methylimidazole (ZIF-67 growth solution) followed by aging for 24 h.
- Other supporting matrices used to make aerogel by in-situ method: reduced graphene oxide, wood, and carbon nitride (C_3N_4), etc.

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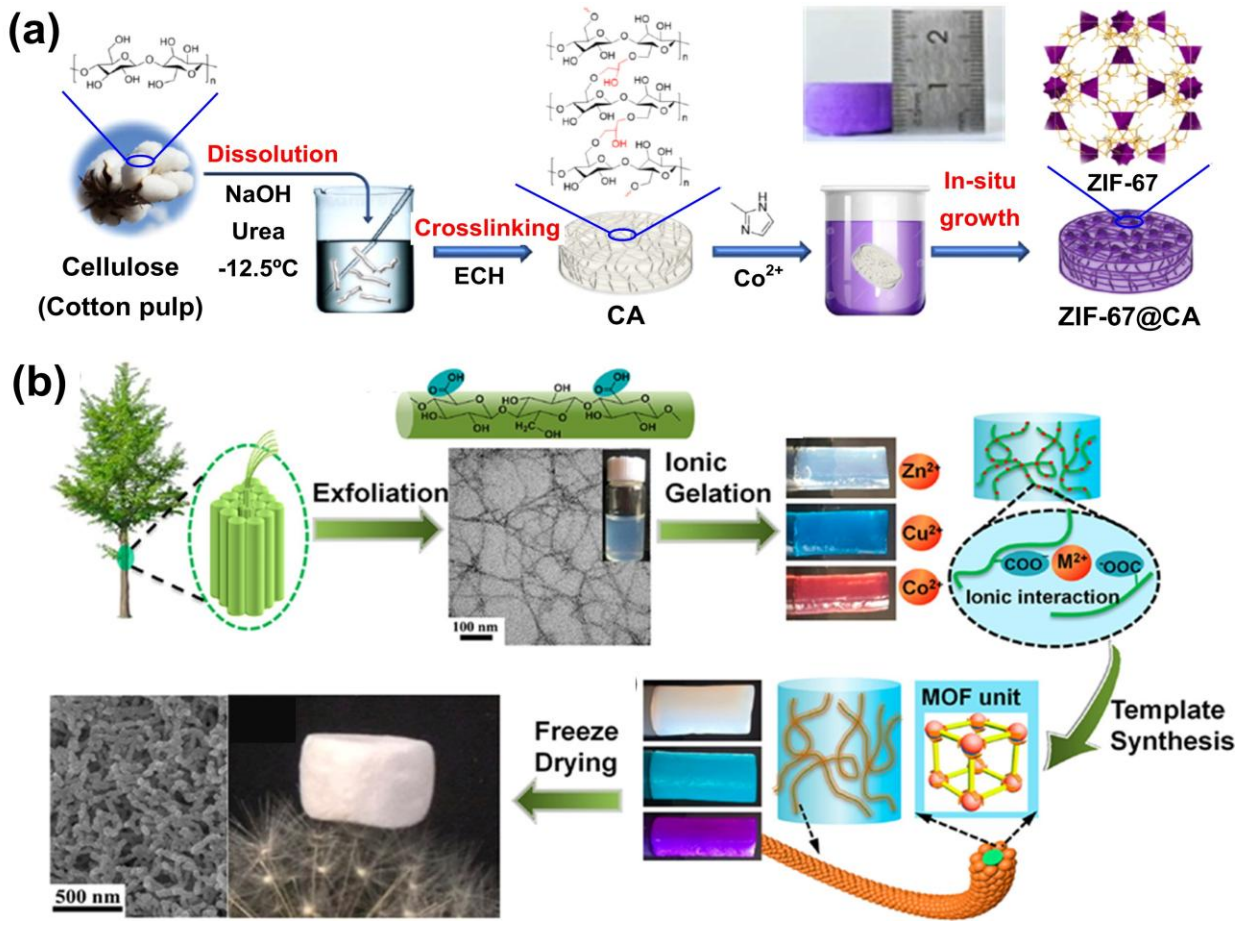
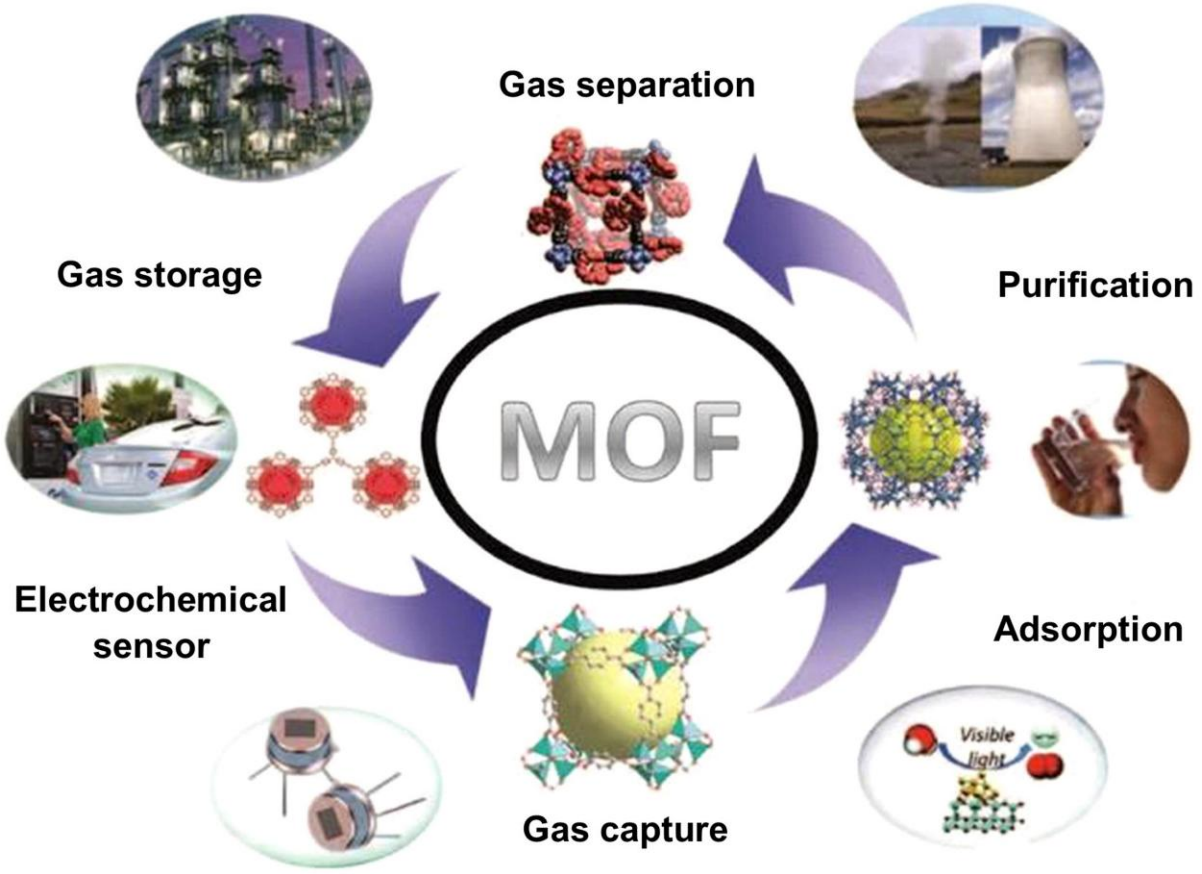


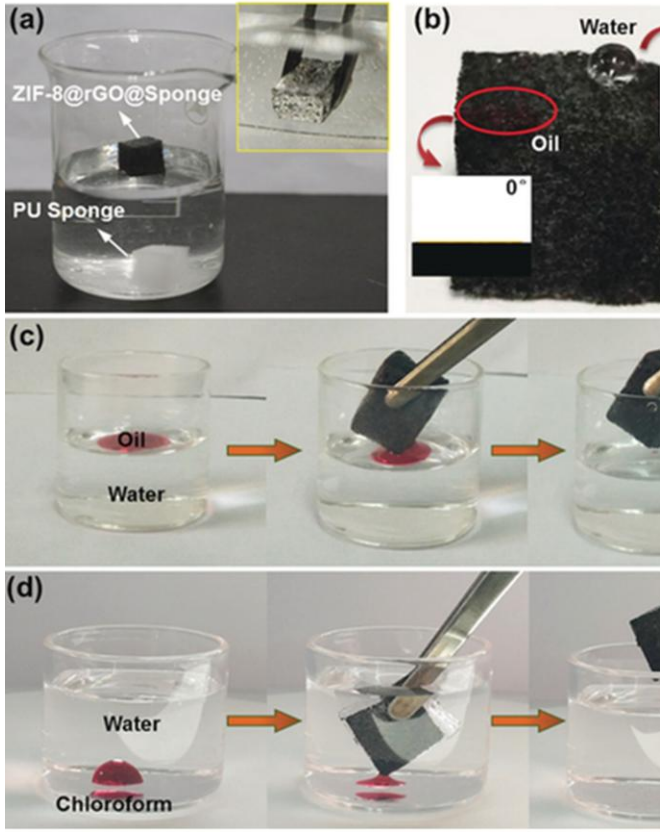
Figure 34. (a) Fabrication of ZIF-67@CA aerogel by in-situ growth of ZIF-67 on cotton pulp-derived cellulose. (b) Fabrication of different MOF aerogels by in-situ growth of different MOFs on cellulose nanofibrils derived from dry kraft pulp.

3.11. APPLICATIONS OF MOFs

- Antibacterial agent
- Oil adsorption
- Dye degradation (adsorption)
- Electrochemical sensor
- Gas capture/separation
- Gas storage



3.12. MOF FOR OIL ADSORPTION



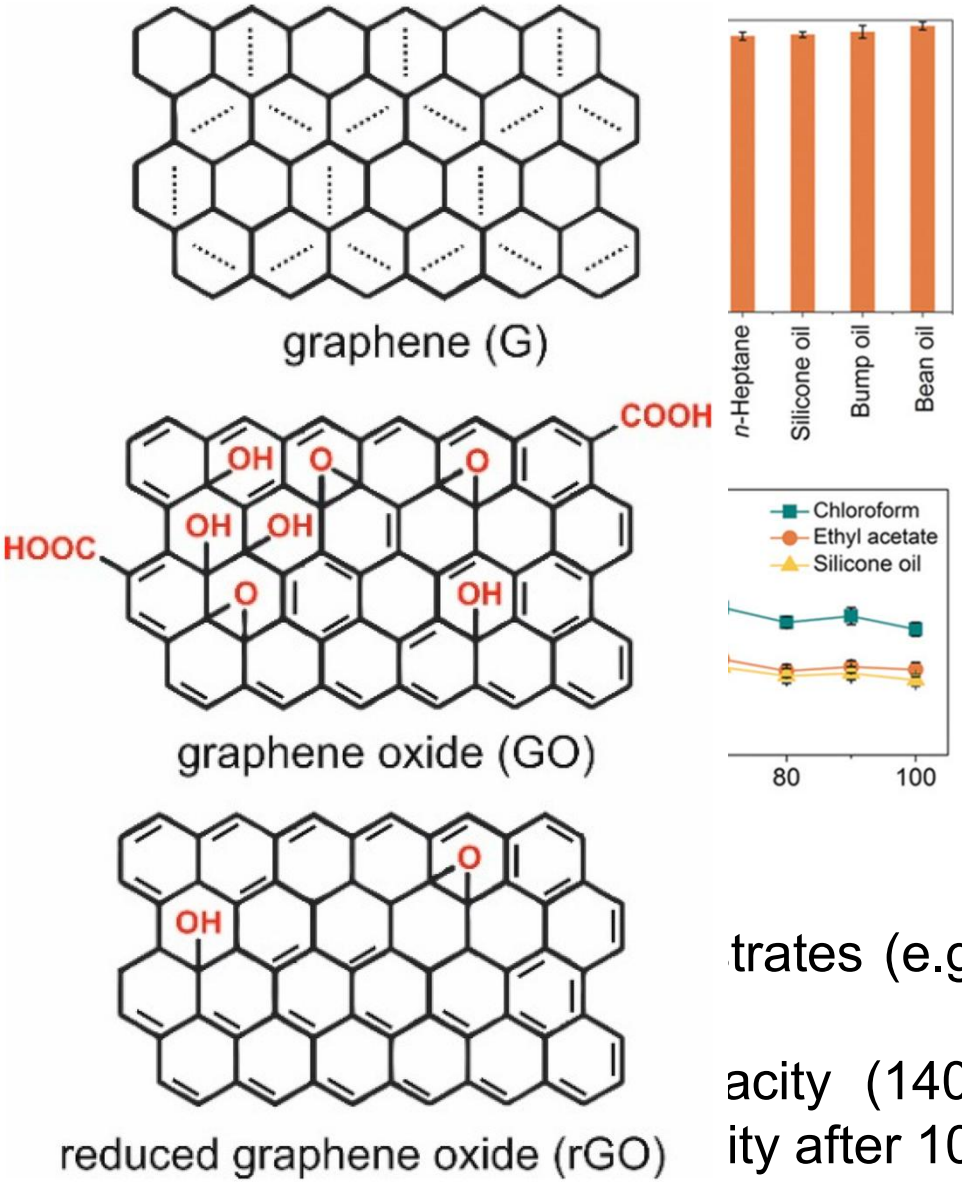


Figure 35. (a) Photo of pure PU and ZIF-8@rGO@PU sponge in water. (b) Photo depicting oil and water droplets on ZIF-8@rGO@PU sponge. Photos showing the adsorption of dyed silicone oil (c) and chloroform (d) from water. (e) Absorption capacities toward oils and organic solvents. (f) Water-oil separation efficiency. (g) Absorption recyclability test of the ZIF-8@rGO@sponge.

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- MOFs, such as ZIF-8 can… trates (e.g., foams, papers, etc.) and used to absorb oil and other organ…
- ZIF-8@rGO@PU sponge… acity (1400-2900 wt.%) for oil and organic solvents, efficient oil-wate… ity after 100 cycles.
- High absorption capacity → mesoporosity and superhydrophobicity.

3.13. MOF FOR DYE ADSORPTION

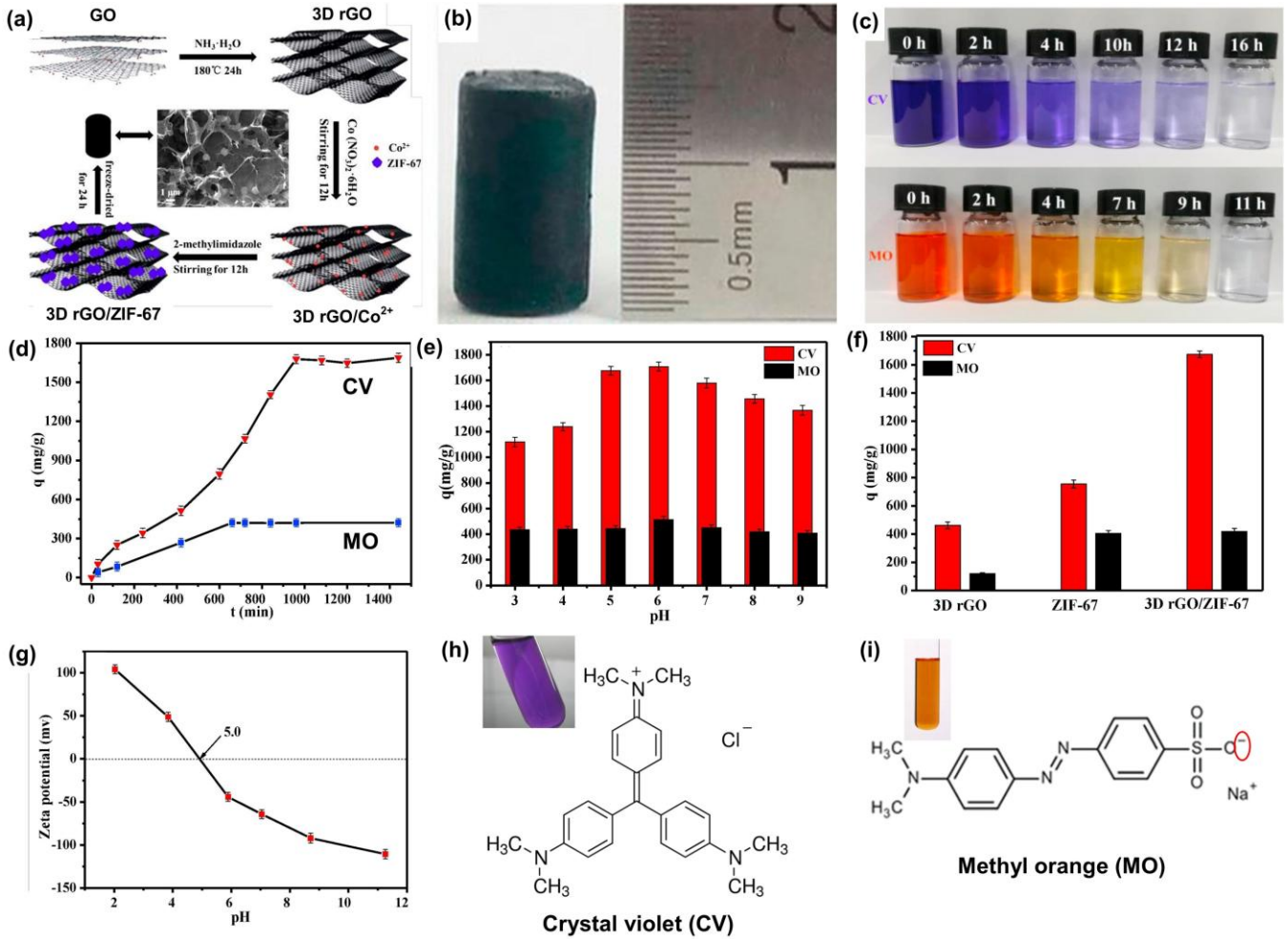


Figure 36. (a) Preparation process and (b) digital photo of reduced graphene oxide (rGO)/ZIF-67 aerogel. (c) Digital photo of crystal violet (CV) and methyl orange (MO) solution at different adsorption time. Effects of (d) contact time and (e) pH on CV and MO adsorption capacity. (f) Adsorption capacities of 3D rGO, ZIF-67 and 3D rGO/ZIF-67 aerogel for CV and MO. (g) Zeta potential of 3D rGO/ZIF-67 aerogel. Chemical structures of (h) CV and (i) MO.

- Chemical interactions between benzene rings in rGO/ZIF-67 and CV & MO by π-π interactions.
- rGO/ZIF-67 shows higher adsorption capacity for CV than MO.

- CV is a positively charged (“cationic”) dye and MO is a negatively charged (“anionic”) dye.
- At pH 6, the surface charge of rGO/ZIF-67 aerogel is negative → promotes <u>electrostatic adsorption with CV but not MO</u>.

3.14. DYES AND MOF ADSORBENTS

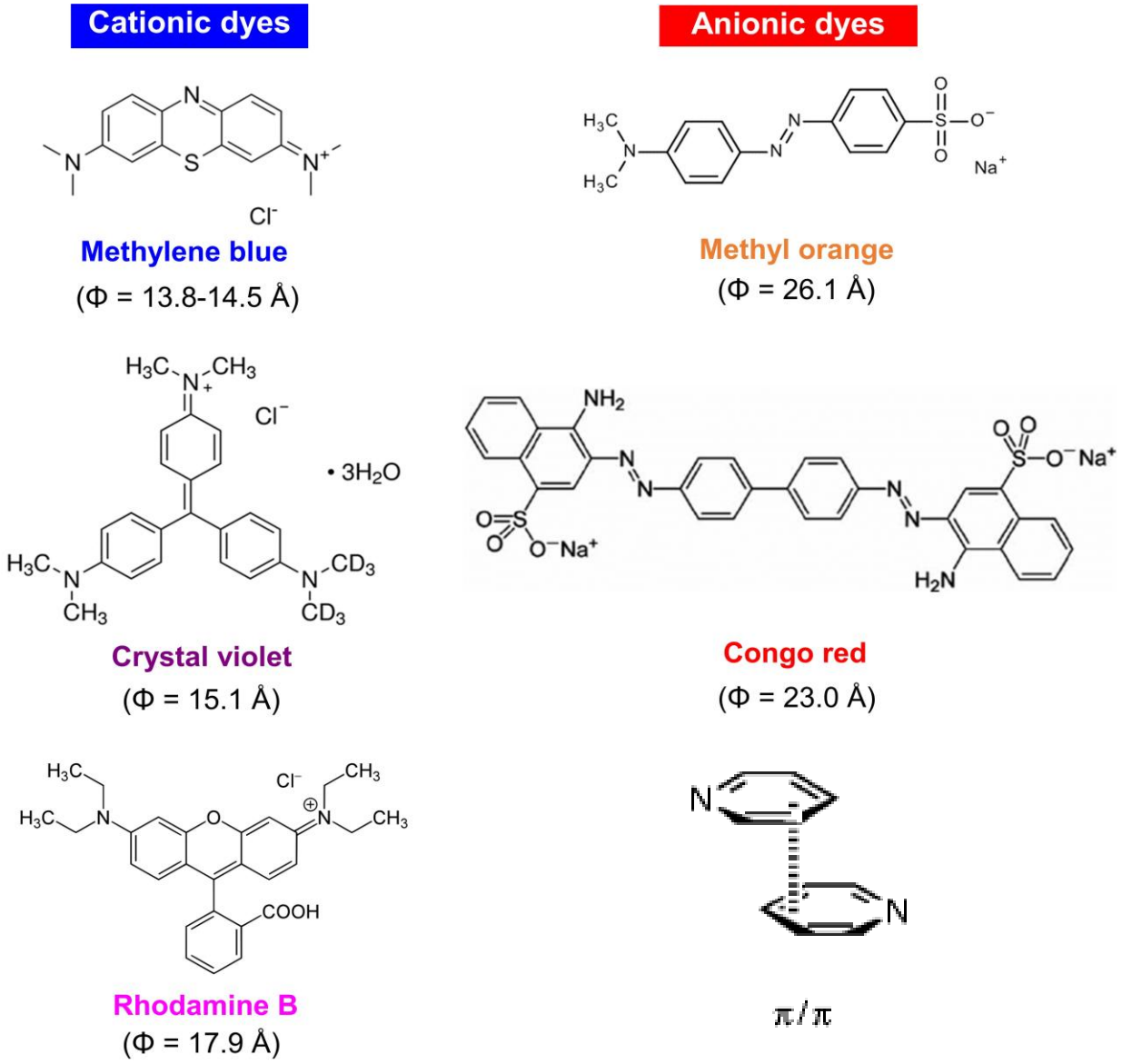


Figure 37. Examples of cationic and anionic dyes.

- Three important factors to consider: surface charge, pore size, and π-π interactions.
- **Surface charge:** anionic dye → MOF with positively charged surface & cationic dye → MOF with negatively charged surface.
- **Pore size:** pore diameter > dye molecule size → to enable effective adsorption of dye inside the pores of MOF.
- **π-π interactions:** desirable to promote chemical (non-covalent) interactions between the aromatic groups in dye and MOF structure.

3.15. OPTICAL PROPERTIES BY UV-VIS

- **UV-visible spectroscopy:** to obtain the absorbance spectra of a compound in solution (e.g., dye solution) or as a solid (visible range: 200-800 nm)

- **Components in UV-vis spectrophotometer:** Light source, monochromator, sample, and detector.

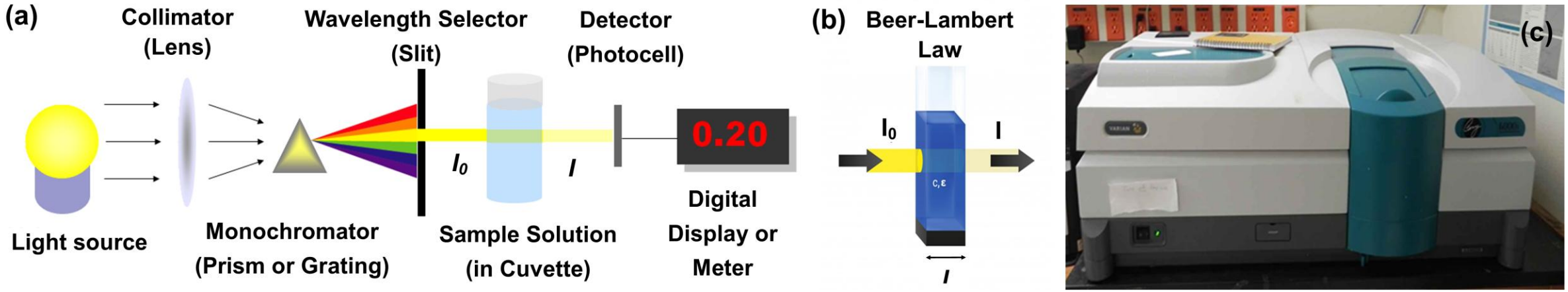


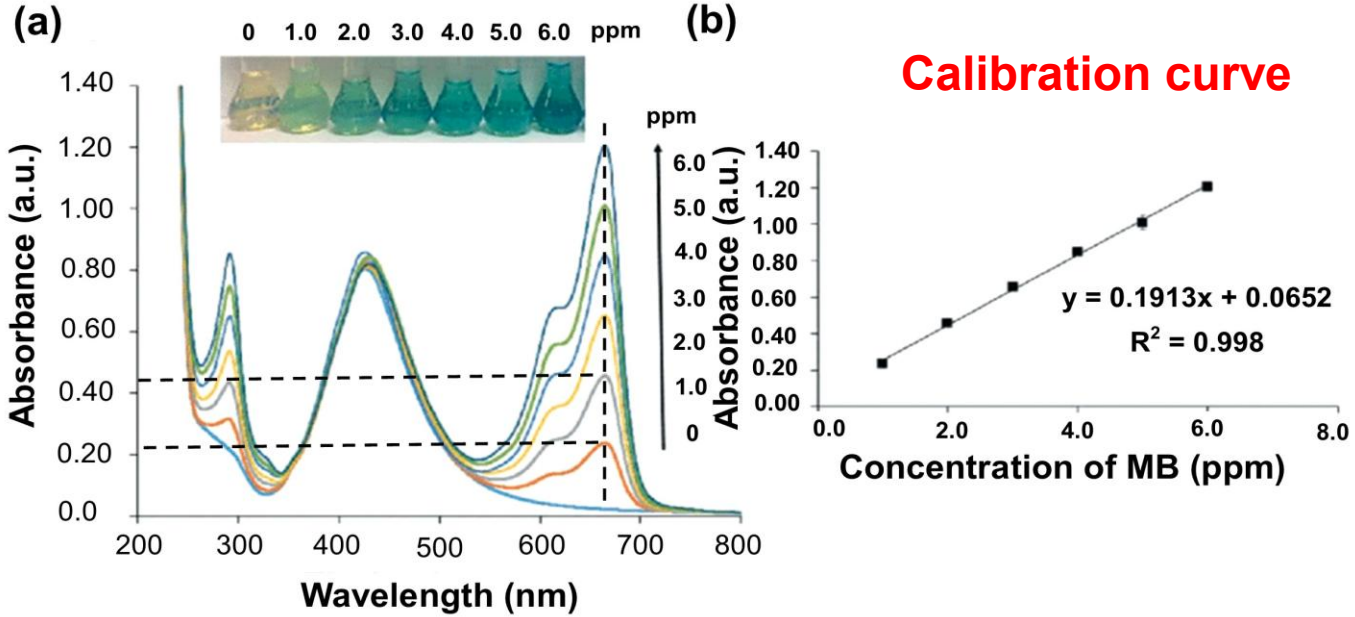
Figure 38. (a) Typical components of a UV-vis spectrophotometer. (b) Beer-Lambert Law. (c) Digital photograph of a UV-vis spectrophotometer.

3.16. INTERPRETING UV-VIS RESULTS

- **Beer-Lambert Law:** *"The amount of absorbed light (absorbance) is proportional to the solution concentration."*

$$A = \varepsilon l C$$

where A is absorbance (unitless or arbitrary units), ε is the molar absorption coefficient ($M^{-1}\,cm^{-1}$), l is the path length of the cuvette (usually 1 cm), and C is the solution concentration (M).



$$A = 0.1913C + 0.0652$$

$$C = A - 0.0652/0.1913$$

Figure 39. (a) UV-vis spectra of methylene blue (MB) solutions with different concentrations and (b) the corresponding concentration *versus* absorbance plot, showing a linear correlation.

3.17. ADSORPTION ISOTHERMS

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❑ **Adsorption isotherms:**

- Describe the amount of substance adsorbed on the adsorbent surface as a function of the concentration of the substance in gas phase or solution.
- Give information about **the type of adsorption (monolayer or multilayer)**

❑ Two common adsorption isotherms:

(1) Langmuir isotherm: assumes that the adsorbent surface is homogeneous, and all binding sites have the same affinity for the adsorbate, leading to monolayer adsorption.

$$\frac{C_e}{q_e} = \frac{1}{q_m K_l} + \left(\frac{1}{q_m}\right) C_e$$

y = b + ax

where C_e is adsorbate concentration at equilibrium. q_e and q_m (mg/g) are the adsorption capacity at equilibrium and maximum adsorption capacity, respectively; K_l is the Langmuir constant.

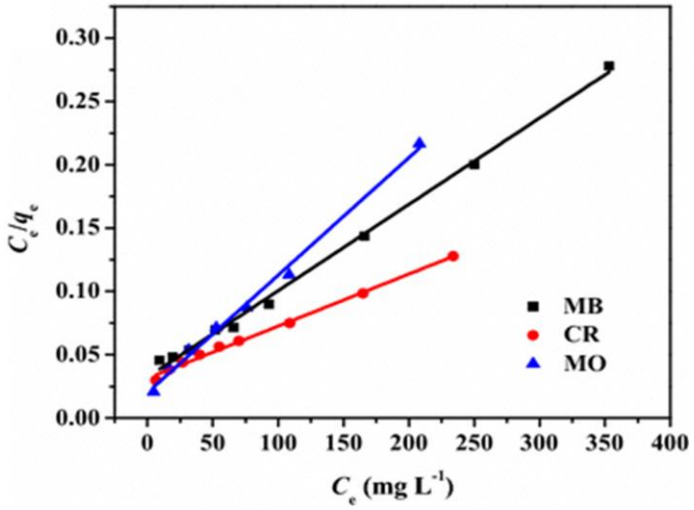


Figure 40. Langmuir isotherms of dye adsorption.

3.17. ADSORPTION ISOTHERMS

(2) Freundlich isotherm: assumes that the adsorbent surface is heterogeneous, and different binding sites have varying affinity for the adsorbate, leading to multilayer adsorption.

$$\log (q_e) = \log (K_f) + (1/n) \log (C_e)$$

$$y \quad = \quad b \quad + \quad ax$$

where C_e is adsorbate concentration at equilibrium. q_e is the adsorption capacity at equilibrium, and K_f and 1/n are Freundlich constants.

$$q_e = \frac{(C_0 - C_e) \times V}{m}$$

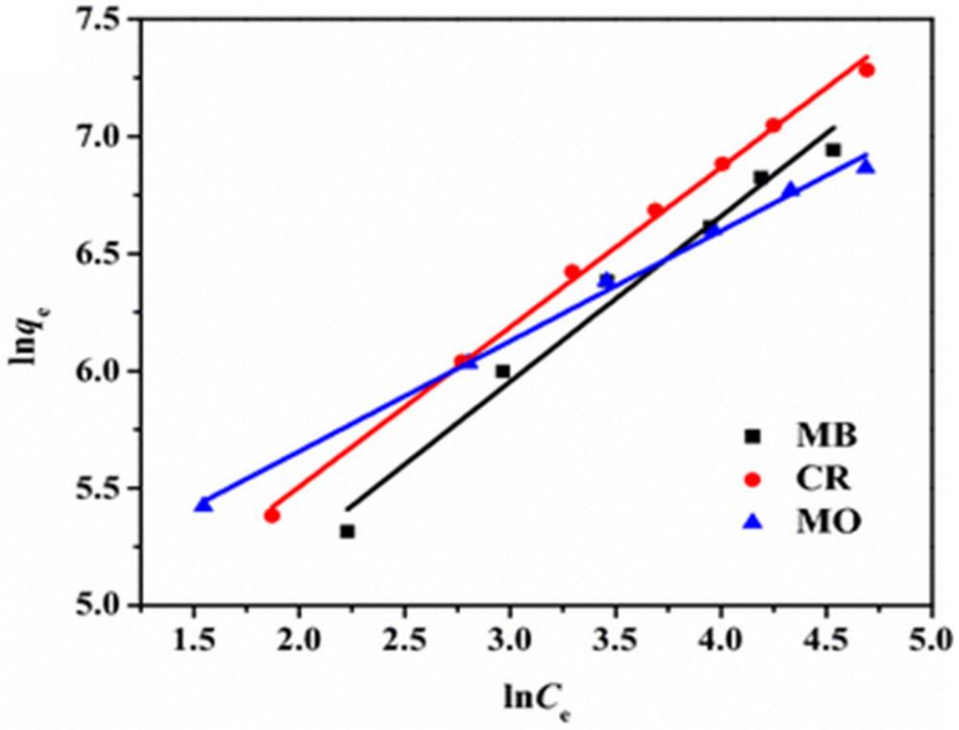


Figure 41. Freundlich isotherms of dye adsorption.

3.17. ADSORPTION ISOTHERMS

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Criteria	Langmuir	Freundlich
Adsorbent surface		
Binding sites		
Type of adsorption		

3.18. ADSORPTION KINETICS

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- **Adsorption kinetics:** useful for predicting the adsorption mechanism and rate-limiting step.

- Two common models to describe the adsorption kinetics for dye adsorption:

(1) Pseudo first-order model: assumes that *physisorption is the rate-limiting step*. Hence, the *adsorption rate depends on adsorbate concentration*. More suitable for describing initial adsorption.

$$\log(q_e - q_t) = \log(q_e) - \frac{k_1}{2.303}t$$

$$y = b + ax$$

where q_e and q_t (mg/g) are the adsorption capacity at equilibrium and time t, respectively; k_1 is the equilibrium rate constant (L/min) in the pseudo first-order model.

3.18. ADSORPTION KINETICS

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(2) Pseudo second-order model: based on the assumption that *the rate-limiting step is chemical adsorption.* In this case, *the adsorption rate depends on the adsorption capacity rather than the concentration of the adsorbate*. Predicts the behavior in the entire adsorption range.

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \left(\frac{1}{q_e}\right)t$$

$$y = b + ax$$

where q_e (unitless) and q_t (mg/g) are the adsorption capacity at equilibrium and time t, respectively, and k_2 is the equilibrium rate constant of the pseudo second-order model (g mg^{-1} min^{-1}).

- Rearranging the above equation to get an equation describing q_t as a function of t and then plot q_t versus t

$$q_t = \frac{k_2 q_e^2 t}{1 + k_2 q_e t}$$

3.18. ADSORPTION KINETICS

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Criteria	Pseudo 1st-order	Pseudo 2nd-order
Rate-limiting step		
Adsorption rate		
Behaviour prediction		

3.18. ADSORPTION KINETICS

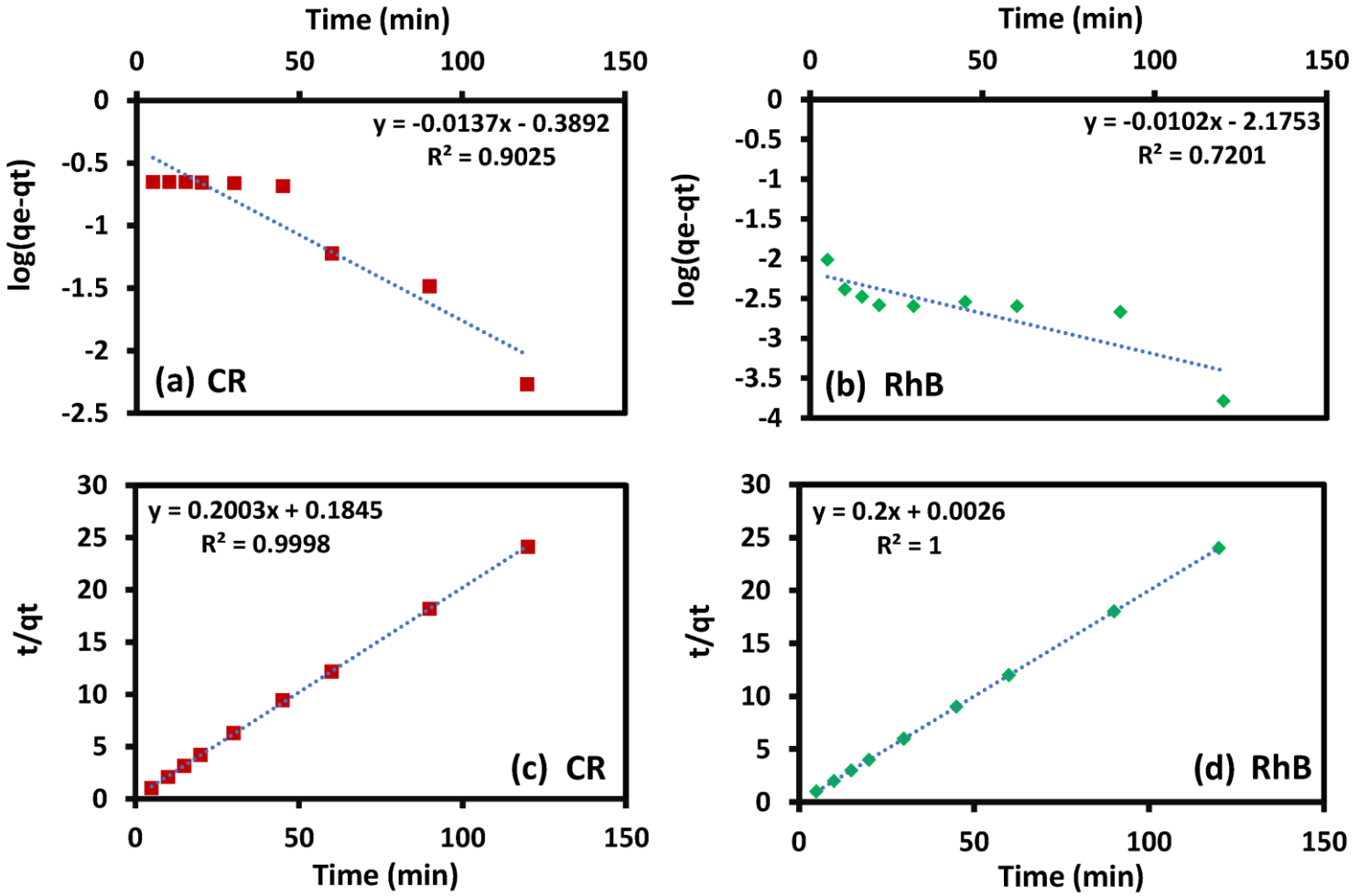


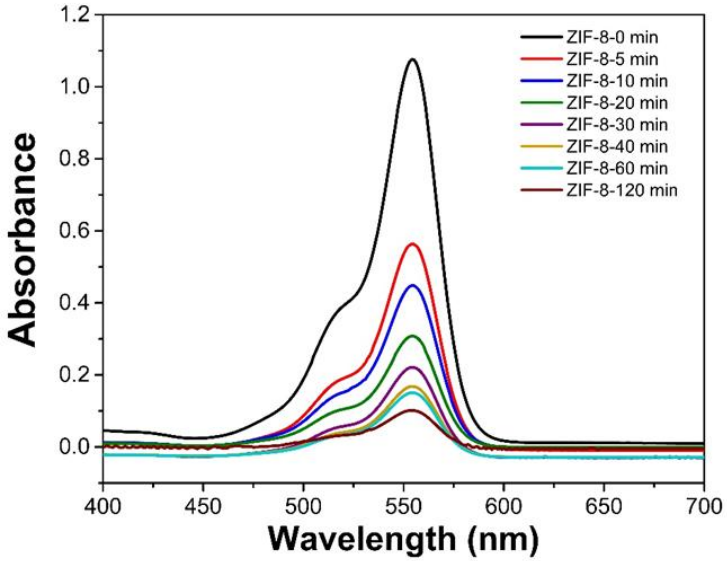
Figure 42. Adsorption kinetics: pseudo-first-order model of (a) CR dye; (b) RhB dye and pseudo-second-order model for the adsorption of (c) CR dye; (d) RhB dye (T= 298 K, pH = 2.0, Co = 25 mg L^{-1}, mass = 0.15 g, V = 20 mL).

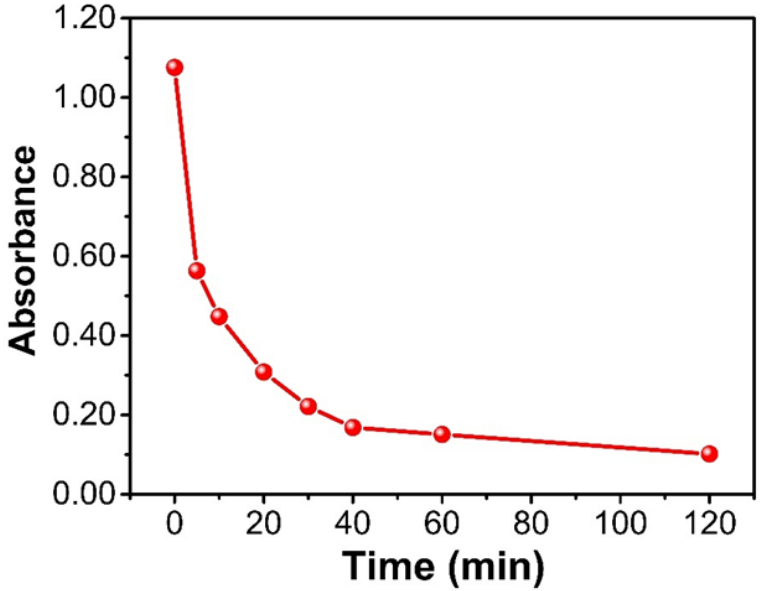
- **Pseudo first-order:** Plot t *versus* log (q_e-q_t).
- **Pseudo second-order:** Plot t/qt *versus* t.
- Then compare the **R^2 value** to determine which model is more suitable to describe the adsorption kinetics of a particular dye by the adsorbent.
- **Closer R^2 value to 1** indicates higher suitability.

QUESTIONS?

QUESTION 1

Tom employed a ZIF-8 aerogel to absorb or remove Rhodamine B dye contaminant from water. He measured the absorbance of the Rhodamine B solution with increasing contact time with the ZIF-8 aerogel from 0 minute to 120 minutes using the UV-vis spectroscopy technique and obtained the following result:

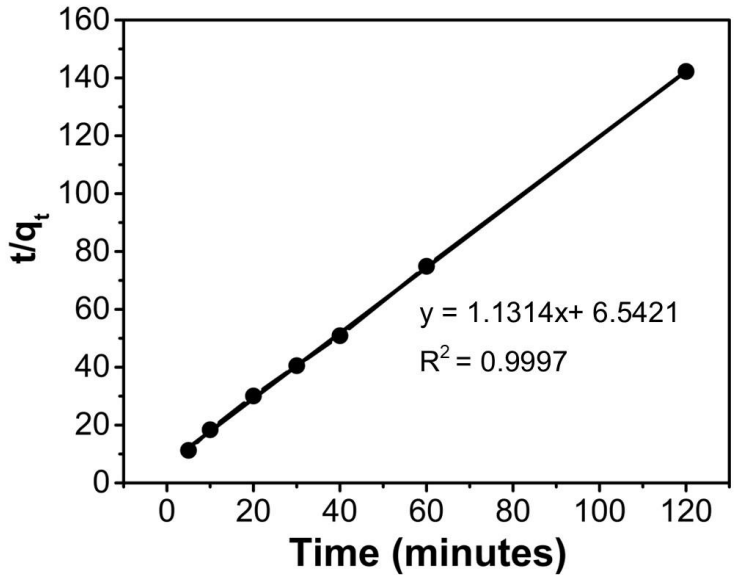




Using the above data, plot the absorbance *versus* time plot for the Rhodamine B dye degradation by the ZIF-8 aerogel.

QUESTION 2

(b) From the above data, Tom plotted the pseudo second-order model to analyse the adsorption kinetics of Rhodamine B dye by the ZIF-8 aerogel and found the model to be highly suitable to describe the adsorption kinetics due to the high R^2 value (almost 1), as shown below:



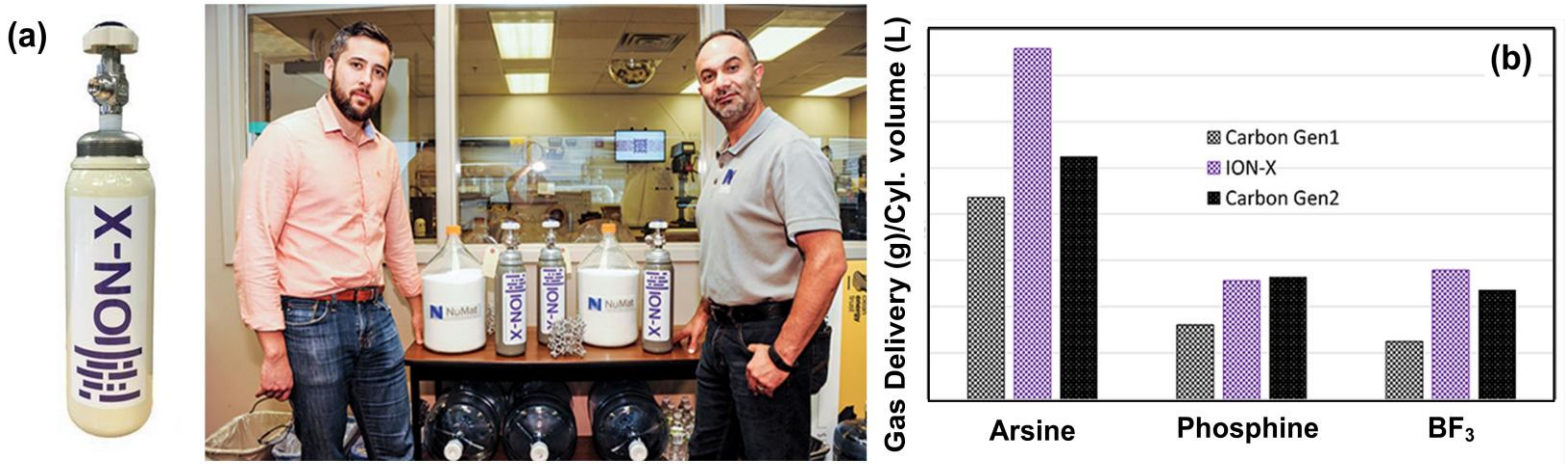
Using the above linear plot, determine the pseudo-second-order rate constant (k_2) and the adsorption capacity at equilibrium (q_e) values (round both values to two decimal places).

Answer: From the linear plot: slope = 1.1314 and intercept = 6.5421

Slope = $\frac{1}{q_e}$ = 1.1314, so $q_e = \frac{1}{1.1314}$ = 0.88 mg/g

Intercept = $\frac{1}{k_2 q_e^2} = \frac{1}{k_2 * (0.88)^2}$ = 6.5421; so $k_2 = \frac{1}{6.5421 * (0.88)^2}$ = 0.20 g/mg/min

COMMERCIAL MOF-BASED STORAGE



Sources:

https://www.chemistryworld.com/news/mofs-offer-safer-toxic-gas-storage-/1017610.article

https://www.numat.tech/ion-x-oem-data

https://www.youtube.com/watch?v=kxoeD8vBmZY

Figure. (a) Digital photograph of ION-X gas cylinder and (b) dopant gas delivery comparison between carbon-based incumbent products and ION-X.

- Commercial ION-X gas cylinder (2.4 L) was developed by a start-up company in U.S., NuMat Technologies.
- Used to store arsine, phosphine & boron trifluoride dopant gases → semiconductor industries.
- ION-X gas cylinder shows higher gas delivery than conventional carbon-based adsorbents → ultrahigh surface area and tailored pore size. Also, safer than conventional gas cylinders → low internal pressure (1 atm).